Removal of the spatial cutoff for weakly coupled $P(\varphi)_2$: norm convergence of the ground-state net on the quasi-local algebra

Part I — unified manuscript

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Repair status (16 August 2026; superseded on 31 October 2026, see below). This is the canonical Part I manuscript. The proof below is an implication from the imports in §2.4; it is not a proof of those imports. In particular, the exact slab-uniform moment estimate U_{GJS} is now stated explicitly in Assumption A8(vi). The printed GJS text gives strong evidence for that reading, but Appendix C is a source-mechanism audit and a proof of analytic prerequisites, not a reconstruction of the GJS cluster expansion. Consequently Theorem 12.2 is conditional on U_{GJS} and the other enumerated imports. The localization omission found in the CE2-to-gap passage and the ultraviolet-mollifier specification are repaired below.

Scope amendment (5 September 2026; superseded on 31 October 2026, see below). The status of U_{GJS} is re-based. It is no longer described as a quantifier strengthened beyond the source: it is the cutoff-uniform quantifier that [7] itself states (p. 629) and consumes in its printed proofs of Theorems 1.1.7 and 1.1.1 (pp. 594–597). Assumption A8(vii′) quotes the passages verbatim and the Scope paragraph of Assumption A8 records the reading. U_{GJS} remains a named hypothesis of Theorem 12.2: it is imported at the source's own scope, not re-derived here. No proof in the manuscript changed.

Re-sourcing of A8(vi) (31 October 2026). Theorem 12.2 is proved from the literal imports A1–A8. Its moment input A8(vi) is GJS Theorem 1.1.11, (1.1.16), at $B = I$, $n = 0$ and any $0 < T < 1$ (p. 596), the case GJS name on p. 629 and specialize to $T = 0$ in their proof of Theorem 1.1.8 (p. 600); the record does not re-derive the cluster expansion. The two paragraphs above are kept for the record: the reading of 5 September 2026 used the same pages as evidence for Theorem 1.1.8 and is the same mathematics with different bookkeeping. No proof in the manuscript changed. The planned derivation of the local moment from Theorem 1.1.7 alone, and the one case of Theorem 1.1.7 that its printed proof does not reach (Scope paragraph of Assumption A8), are recorded in `ugjs/PLAN.md`.

Abstract

Let $H(g)$ be the spatially cutoff $P(\varphi)_2$ Hamiltonian, Ω_g its ground state, and ω_g the induced state on the time-zero quasi-local algebra $\mathfrak{A}$. Glimm and Jaffe (1970) constructed an infinite-volume vacuum as a w^* *limit point* of space-averaged ω_g; the cited construction does not prove convergence of the net itself. Guerra–Rosen–Simon (1975) formulated analogous Euclidean local-convergence questions as their Problems 1–2. We prove: in the Glimm–Jaffe–Spencer weak-coupling region $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, the *full*

net (ω_g) — no subnets, no averaging — converges in norm on every local algebra,

$$\lim_{g\in\mathcal{G}} \big\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\big\| = 0,$$

with an explicit super-polynomial rate in $\operatorname{dist}(O, \{g < 1\})$; the limit is locally normal, translation invariant, $\mathbb{Z}_2$-invariant for even $\mathscr{P}$, and is a ground state of the Glimm–Jaffe dynamics $(\mathfrak{A}, \alpha)$ in the spectral sense. The proof is Hamiltonian: a local-perturbations-perturb-locally estimate built on an exact spectral filter and the exact light cone makes the net Cauchy; its two hypotheses — a cutoff-uniform gap spec $H(g) \subseteq \{0\} \cup [m_1, \infty)$ and a uniform local interaction moment — are derived from two printed theorems of Glimm–Jaffe–Spencer (Ann. of Math. 100 (1974)): the exponential-clustering Theorem 1.1.7 and the cutoff-uniform moment estimate of Theorem 1.1.11, (1.1.16), "uniformly in h", in the specialization $B = I$, $n = 0$ (pp. 600 and 629) that the source itself prints as its proof of Theorem 1.1.8. Both are imported as printed; neither is re-proved. The proof also imports, and explicitly identifies in §2, the standard cutoff-construction results on self-adjointness and the simple positive ground state, Feynman–Kac–Nelson transfer, finite-volume stability and hypercontractivity, domain smoothing and a common core, finite propagation and patching, and covariance. No Euclidean uniqueness theorem, global Markov property for the infinite-volume measure, or reverse Osterwalder–Schrader reconstruction is used. We do not claim uniqueness of ω_∞ among all locally normal ground states, a sharp exponential rate, or anything outside the weak-coupling region.

Contents

1 Introduction **3**
- 1.1 The problem . . . 3
- 1.2 The result . . . 3
- 1.3 What is not claimed . . . 4
- 1.4 The proof strategy in operator-algebraic terms . . . 4

2 Setting, notation, and standing imports **6**
- 2.1 The model . . . 6
- 2.2 The three cutoff classes . . . 7
- 2.3 Notation freeze . . . 7
- 2.4 Standing imports . . . 7
- 2.5 Immediate consequences of the imports . . . 14
- 2.6 Standing weak-coupling assumption . . . 14

3 Dimensionless weak-coupling reduction **15**

4 Slab transfer **17**

5 The uniform local interaction moment **18**

6 GJS norms of sharp-time polynomial observables **19**

7 The uniform gap **20**

8 Affine-path regularity **24**

9 Filter, spectral separation, and clustering **26**

10 The one-path LPPL estimate **28**

11 Convergence of the full net; local normality **30**

12 Ground state, invariances, and the Main Theorem **31**

13 The two-phase dichotomy **33**

14 Discussion **34**

A Constants **35**

B Provenance, corrections history, and dossier **36**

C h-uniformity: source audit and analytic prerequisites **37**

1 Introduction

1.1 The problem

The $\lambda\varphi_2^4$ (more generally $P(\varphi)_2$) model was constructed by Glimm and Jaffe in the series I–IV [1, 2, 3, 4]. Paper III builds the physical vacuum as follows: the finite-volume ground states Ω_g of the cutoff Hamiltonians $H(g)$ induce states ω_g on the quasi-local algebra $\mathfrak{A}$; after a space-translation averaging, a w^*-compactness argument produces a *limit point* ω, which is shown to be locally Fock [3, Thms. 2.1–2.3]. Whether the net (ω_g) *converges* — so that the vacuum is canonical rather than selected by a limit procedure — was not proved there; the Expositions volume states plainly that the $g \to 1$ limit of the vacuum expectation values "has not been proved" [6, §4]. Guerra, Rosen and Simon list analogous *Euclidean* local L^p and finite-volume Gibbs-state convergence statements as Problems 1–2 of [10, pp. 125–126]; their note added on p. 128 records Newman's solution of the small-coupling $p = 1$ versions. (Some later surveys state the Hamiltonian cutoff limit as a convergence; the cited primary construction itself is unambiguously a limit-point construction.)

On the Euclidean side, convergence of *Schwinger functions* at weak coupling follows from the cluster expansion of Glimm–Jaffe–Spencer [7]. Albeverio–Høegh-Krohn–Zegarlinski prove Euclidean Gibbs uniqueness under their stated hypotheses [13]; Bauerschmidt–Dagallier–Weber prove uniqueness of the invariant measure for the Φ_2^4 stochastic-quantization equation above the critical temperature [14]. Neither statement is, by itself, a theorem about norm convergence of this Hamiltonian ground-state cutoff net. The cited primary construction does not prove that local-norm convergence, and no such theorem is identified in the references reviewed here. The Hamiltonian-renormalisation programme of Rodriguez Zarate–Thiemann cited here is explicitly restricted to finite volume [15].

1.2 The result

> *In the GJS weak-coupling region the full net $(\omega_g)_{g\in\mathcal{G}}$ converges in norm on every local algebra; the limit is locally normal, translation and (for even $\mathscr{P}$) $\mathbb{Z}_2$ invariant, and is a ground state of the Glimm–Jaffe dynamics. An explicit rate holds: $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ for $g \equiv 1$ on O_d, with Ψ_{m_1} decaying faster than every inverse power.*

See Theorem 12.2. The logical structure is deliberately Hamiltonian and compactness-free:

step	content	where proved
(i)	GJS CE2 and U_{GJS} (Thm 1.1.11), both printed	Assumption A8 (§2)
(ii)	uniform local moment (M_2)	§5
(iii)	uniform gap (G): $\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty)$	§7
(iv)	LPPL/Cauchy: (G)+(M_2) $\Rightarrow$ full-net convergence + rate	§§8–11
(v)	the limit is an α-ground state; invariances	§12

Steps (ii)–(iii) convert Euclidean input to Hamiltonian statements through slab Feynman–Kac–Nelson transfers in which every limit is taken at fixed g; no uniform gap is presupposed anywhere in their proofs (the gap of step (iii) is an output). Step (iv) replaces the compactness of [3] by a Cauchy argument: the same gap that yields clustering also yields a local-perturbations-perturb-locally estimate, which bounds $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\|$ whenever both cutoffs equal 1 on O_d. Step (v) is an exact spectral identity: the ground-state property of the limit costs an identity, not an estimate.

The dependency graph, as established by the independent audit of the predecessor manuscript (Appendix B), is:

$$\begin{array}{c}
\text{cutoff self-adjointness, simple positive ground state, FKN, domains} \\
+ \quad \text{GJS CE2, } U_{\mathrm{GJS}} \text{ (Thm 1.1.11), scaling}
\end{array} \Longrightarrow \begin{array}{c}(\mathrm{M}_2) \text{ uniform local moment} \\ (\mathrm{G}) \text{ uniform cutoff gap}\end{array}$$

$$(\mathrm{M}_2) + (\mathrm{G}) + \text{finite propagation} + \text{patching} \\ +\text{affiliation} + \text{path regularity} \Longrightarrow \text{LPPL estimate} \Longrightarrow \begin{array}{c}\text{full-net local-norm convergence} \\ \text{local normality}\end{array}$$

$$\text{local-norm convergence} + \text{patching} + \text{covariance} \Longrightarrow \begin{array}{c}\text{spectral ground state} \\ \text{translation and parity invariance.}\end{array}$$

1.3 What is not claimed

(a) *Uniqueness*: Theorem 12.2 identifies the limit of *this* cutoff family; it does not show that every locally normal ground state of $(\mathfrak{A}, \alpha)$ equals ω_∞. (b) *Sharp rate*: in the exactly solvable quadratic model the true rate is $\mathrm{poly}(d)\, e^{-2m_1 d}$ (companion note `m2/`); Ψ_{m_1} is super-polynomial but not exponential, and this is intrinsic to the filter method used. (c) *Beyond weak coupling*: under the explicit phase-classification hypotheses stated in Hypothesis 13.1, a cutoff-uniform gap is incompatible with that two-phase structure (Theorem 13.2). Thus an extension to such a broken phase would have to abandon the uniform-gap input; no claim is made outside $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. (d) *Wavelet uniformity*: uniformity in a lattice/wavelet resolution is a separate problem (Part II of this programme) and is not addressed here.

1.4 The proof strategy in operator-algebraic terms

The topology, and why it is the right one. Each ω_g is the vector state of the simple positive ground vector Ω_g of $H(g)$, restricted to the quasi-local algebra $\mathfrak{A}$; the cutoffs form a directed set under inclusion of plateaus. The classical construction [3] produced a vacuum as a w^* *limit point* of space-averaged ω_g: compactness selects a state but proves neither convergence nor canonicity. Theorem 12.2 asserts convergence of the full, unaveraged net in the norm of every local dual $\mathfrak{A}(O)^*$ — the strongest locally uniform statement available — and Corollary 12.3 then makes the historical construction subsequence- and averaging-independent. Norm convergence on local duals also transports normality: the limit lies in the local Fock folia (Proposition 11.2).

Why standard tools fail. The difference $V(g') - V(g)$ has infinite norm, so the quasi-adiabatic/automorphic-equivalence machinery for gapped lattice systems [16, 17] does not start; ergodicity-type arguments give extremality of limit points, not uniqueness; and Euclidean convergence of Schwinger functions lives in the wrong topology — there is no soft

passage from moments to norms on local von Neumann algebras. The architecture here is instead a *Cauchy* argument built from a local-perturbations-perturb-locally estimate: if $g \equiv g' \equiv 1$ on O_d, then $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ with Ψ_{m_1} super-polynomial (Proposition 10.4, Lemma 10.5). Plateaus are cofinal and local duals are complete, so the net converges (Theorem 11.1) — no compactness anywhere.

Pillar 1: two cutoff-uniform Hamiltonian estimates from Euclidean input. The Cauchy estimate consumes exactly two nonstandard inputs, both uniform over the whole Hamiltonian cutoff class $\mathcal{C}_{\mathrm{H}}$: the gap (G) of Theorem 7.8, and the local moment (M_2) of Theorem 5.1. Both are transferred from the two cluster-expansion estimates of [7] for the interacting Euclidean measures dq_h — the cutoff-uniform moment bound (U_{GJS}) (Assumption A8(vi), the printed Theorem 1.1.11 of [7] at $B = I$, $n = 0$), and the exponential clustering of Source Fact 2.2. Appendix C records the source mechanism and proves the normalization and stability prerequisites; it does not replace the source's polymer expansion. The transfer device is the Feynman–Kac–Nelson slab formula: $\psi_T = e^{-TH(g)}\Omega_0$ normalizes to the ground state, and sharp-time Euclidean moments become matrix elements of $e^{-tH(g)}$. One structural point deserves emphasis: $\hat{\psi}_T \to \Omega_g$ uses *simplicity only* (Lemma 4.4) — no gap is presupposed, and the gap of Theorem 7.8 is an output, so there is no circularity. Converting semigroup decay on centered polynomial vectors into a spectral gap requires that these vectors be total; this is the density theorem (Theorem 7.7), proved from the factorial moments supplied by (U_{GJS}) (Lemma 7.1) via exponential integrability, strip analyticity of characteristic functionals, Fourier uniqueness, and martingale convergence.

Pillar 2: the derivative estimate. Along the affine interpolation $g_s = (1-s)g + sg'$, first-order perturbation theory for the simple ground state gives $\dot{\Omega}_s = -H_s^{-1} Q_s V(v) \Omega_s$ (Proposition 8.2). Three moves control this, each with an operator-algebraic flavor. *(i) An exact inverse-energy filter* (Lemma 9.1) writes $H^{-1}(1-P_0) = \int W_\gamma(t)\, e^{itH}(1-P_0)\, dt$, converting the reduced resolvent into dynamics. The kernel W_γ decays faster than every inverse power but *cannot* decay exponentially: a kernel with exponential tails has a transform analytic in a strip, which cannot vanish on a real neighbourhood of 0 — as $\widehat{W}_\gamma$ must — without vanishing identically. This is why the final rate is super-polynomial rather than exponential; it is intrinsic to the method (cf. §14). *(ii) The exact light cone*: e^{itH_s} turns $A \in \mathfrak{A}(O)$ into $\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$ with speed ≤ 1 exactly (Proposition 2.7); no Lieb–Robinson theory is needed. *(iii) Clustering with one unbounded local argument* (Lemma 9.3): the correlation between a far perturbation cell V_j and the localized $\beta_t(A)$ is bounded by $2\|V_j\Omega_s\|\|A\|e^{-\gamma R/2}$. The mechanism is a quantitative substitute for the strip-analyticity arguments familiar from positive-energy representations: the two boundary functions have spectral supports in $[\gamma, \infty)$ and $(-\infty, -\gamma]$, locality makes them agree on the interval $|t| < R$, and an optimized Gaussian separator (Lemma 9.2) bounds the value at $t = 0$. Note what the moment (M_2) is doing: the perturbation is unbounded, but only V_j *evaluated in the ground state* enters — the uniform moment is the operator-algebraic softening of the unboundedness. A unit-scale partition of the perturbation, the two-regime split of each time integral, and the count of at most eight cells per distance shell assemble the estimate (Lemma 10.1 through Proposition 10.4).

Assembly and identification. Cauchy-ness in each $\mathfrak{A}(O)^*$ gives compatible limit functionals and the state ω_∞; local normality follows by a monotone-net argument (Proposition 11.2). The ground-state property costs an identity, not an estimate: for each cutoff and $|t| < d(g, O)$, patching (P) makes $\omega_g(A\alpha_t(B))$ literally the Fourier–Stieltjes transform of a measure supported in $[0, \infty)$, and the positive-energy condition passes to the limit by locally uniform convergence (Proposition 12.1). Translation invariance needs no averaging: $g \mapsto \tau_a g$ is an order isomorphism of the cutoff net with cofinal range, so covariance (T) plus net convergence suffices; parity likewise.

Boundaries. All of this sits in the weak-coupling region through the explicit import ledger

of §2.4 and the coordinate convention of Remark 3.3. The theorem does not identify ω_∞ among *all* locally normal ground states of $(\mathfrak{A}, \alpha)$, does not reach a sharp exponential rate (the filter obstruction above; the solvable quadratic model has $\mathrm{poly}(d)\, e^{-2m_1 d}$), and claims nothing beyond weak coupling; there, Theorem 13.2 records the expected mechanism — for even $\mathscr{P}$, a cutoff-uniform gap is incompatible with the fully classified two-phase structure of Hypothesis 13.1. In one sentence: *the same uniform gap that gives exponential clustering also gives a local-perturbation stability estimate through an exact inverse-energy filter and the exact light cone; the cluster expansion of [7] supplies the gap and one local moment uniformly in the cutoff; and Cauchy-ness in the local duals replaces every compactness step of the classical construction.*

Provenance and relation to the frozen sources

This document unifies, at full explicitness, the following frozen components of the research record (file pointers in Appendix B): the audited manuscript `m6/main.tex` (v2), the two structured proof reconstructions `audit_2026/lamport_euclidean_to_gap.tex` and `audit_2026/lamport_gap_to_main.tex`, the source-scope and constant gates `m4/r1-check.tex` and `m4/r2-check.tex`, the spectral-passage note `m0/`, and the dichotomy note `m3/`. Every repair identified by the independent audit of 9 August 2026 (register in Appendix B) is incorporated. External inputs are the printed theorems of [7] and the explicitly cited construction results from [1, 2, 5, 11, 12, 10, 6]; they are enumerated once, as Assumptions A1–A8, and no other external mathematical input is used (Corollary 12.3 additionally quotes the averaged-state formula [3, (2.3)–(2.5)] as the definition of the averaged states it identifies).

2 Setting, notation, and standing imports

2.1 The model

$\mathcal{H} = L^2(\mathcal{S}'(\mathbb{R}), d\phi_0)$ is Q-space over the time-zero free field of mass $m_0 > 0$: $d\phi_0$ is Gaussian with covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$. Weyl operators $W(f,h) = \exp i(\varphi_0(f) + \pi_0(h))$ generate the local net

$$\mathfrak{A}(O) = \{W(f,h) : f, h \in C_c^\infty(O) \text{ real}\}'', \qquad \mathfrak{A} = \overline{\textstyle\bigcup_O \mathfrak{A}(O)}^{\,\|\cdot\|},$$

O ranging over bounded open intervals. Let $\mathscr{P}(\xi) = \sum_{n=0}^{2p'} a_n \xi^n$, $a_{2p'} > 0$, $p' \geq 1$, and for real $h \in L^1(\mathbb{R}) \cap L^2(\mathbb{R})$ of compact support

$$V(h) := \lambda \int h(x) :\!\mathscr{P}(\varphi_0(x))\!:_C dx,$$

a self-adjoint multiplication operator in Q-space, affiliated with $\mathfrak{A}(U)$ for every bounded open $U \supseteq \operatorname{supp} h$, and finitely additive on common domains ($V(\sum_j h_j)\xi = \sum_j V(h_j)\xi$ whenever all terms are defined). For a cutoff g as below,

$$K(g) := H_0 + V(g), \quad E(g) := \inf \operatorname{spec} K(g), \quad H(g) := K(g) - E(g),$$

with normalized ground state Ω_g (Assumption A1), $\omega_g := \langle \Omega_g, \cdot\, \Omega_g \rangle$, and $\beta_t^g = \operatorname{Ad} e^{itH(g)}$.

2.2 The three cutoff classes

Definition 2.1 (cutoff classes)**.**

$$\mathcal{C}_{\mathrm{E}} := \{h : \mathbb{R}^2 \to [0,1] \text{ measurable, } \operatorname{supp} h \text{ compact}\} \quad \text{(Euclidean class)},$$
$$\mathcal{C}_{\mathrm{H}} := \{g : \mathbb{R} \to [0,1] \text{ measurable, } \operatorname{supp} g \text{ compact}\} \quad \text{(Hamiltonian class)},$$
$$\mathcal{G} := \{g \in C_c^\infty(\mathbb{R}) : 0 \le g \le 1,\ g \equiv 1 \text{ on a nonempty open interval}\} \quad \text{(plateau net class)}.$$

The first two are the printed classes of [7, p. 589] and [7, (1.6), p. 588], respectively. $\mathcal{G}$ is directed by $g \preccurlyeq g' \iff \{g = 1\}^\circ \subseteq \{g' = 1\}^\circ$. For a bounded interval O and $r \ge 0$ set

$$O_0 := O, \qquad O_r := \{x \in \mathbb{R} : \operatorname{dist}(x, O) < r\} \quad (r > 0), \qquad d(g, O) := \operatorname{dist}\big(O, \mathbb{R}\backslash\{g = 1\}^\circ\big).$$

The roles are fixed once and for all: the Euclidean estimates of Assumption A8 quantify over $\mathcal{C}_{\mathrm{E}}$; the uniform moment and gap of §§5, 7 quantify over $\mathcal{C}_{\mathrm{H}}$; the convergence theorem quantifies over the subnet class $\mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$. The elementary compatibility statements (memberships and affine-path stability) are Lemma 3.1.

2.3 Notation freeze

All later sections conform to the following table; in particular every symbol imported from a frozen source is converted to this notation at transcription time (see `part_i/README.md` for the conversion log).

symbol	meaning / convention
$\mathscr{P}$, $2p'$	interaction polynomial; $\deg \mathscr{P} = 2p'$, $a_{2p'} > 0$, $p' \ge 1$
$\langle \cdot, \cdot \rangle$	inner product, conjugate-linear in the *first* argument
C, μ	time-zero covariance $C = (2\mu)^{-1}$, $\mu = (-\partial_x^2 + m_0^2)^{1/2}$
$\mathcal{C}_{\mathrm{E}}$, $\mathcal{C}_{\mathrm{H}}$, $\mathcal{G}$	cutoff classes of Definition 2.1, in the roles fixed there
$K(g), E(g), H(g), \Omega_g, \omega_g, \beta^g$	cutoff Hamiltonian data (§2.1)
α, τ_a, U_a	patched dynamics (Assumption A6); space translations and their Q-space unitaries
Θ, θ	parity unitary $\Theta = \Gamma(-1)$; parity automorphism $\theta = \operatorname{Ad}\Theta$
K_1, K_2, C_7, C_8, m_1	GJS constants (Assumption A8); normalization $K_2 \ge 1$
$\|\cdot\|_{\mathrm{GJS}}$	printed observable norm (1.1.7)–(1.1.9) of [7]
N_I	number of unit cells $[j, j+1)$ meeting the interval I in positive measure
M	uniform local moment constant of (M_2) (§5)
γ vs. m_1	γ = abstract gap parameter in §§8–10; instantiated $\gamma = m_1$ in §11
$\mathfrak{m}$	total-variation bound in the spectral-separation lemma (§9)
$h_{T,g}$, ψ_s, $\hat{\psi}_s$, c_g	slab objects (Assumption A2; §4)
Ψ_γ	rate function of the one-path estimate (§10)
d_0	LPPL threshold $d_0 = 4 + 2/\gamma$

Two deliberate renamings relative to the frozen sources: the separation-lemma variation bound is $\mathfrak{m}$ (it was M_0 in `lamport_gap_to_main.tex`, colliding with the moment constant M), and $\Theta = \Gamma(-1)$ is reserved for the parity unitary throughout.

2.4 Standing imports

Each import below states (a) the exact statement used, (b) the pinpoint source, (c) a scope remark verifying that the used statement is no stronger than the printed one, (d) the sections that consume it, and (e) the page images documenting it in the audit dossier (`audit_2026/evidence/`, index in `audit_2026/citation_screenshot_dossier.tex`).

Assumption A1 (cutoff construction)**.** For every $g \in \mathcal{C}_{\mathrm{H}}$: $K(g) = H_0 + V(g)$ is essentially self-adjoint and bounded below; its closure (again denoted $K(g)$) has $E(g) = \inf \operatorname{spec} K(g)$ as a *simple* eigenvalue, and the normalized eigenvector Ω_g may be chosen strictly positive $d\phi_0$-a.e. in Q-space. For every real $h \in L^1 \cap L^2$ of compact support, $V(h)$ is self-adjoint on its maximal multiplication domain.

Sources. [7, p. 588]; existence and uniqueness of the cutoff vacuum: [2, Thms. 2.2.1, 2.3.1]; lower bounds and positivity: [5]; the semigroup construction: [11, Thm. 5.3].

Scope. Strict positivity of Ω_g is the standard positivity-improving semigroup argument of the cited construction; simplicity is used below only through the two consequences $c_g := \langle \Omega_g, \Omega_0 \rangle > 0$ and the absence of a second ground vector. *Isolation of $E(g)$ (a gap) is not imported*; the gap of §7 is an output.

Used in. §4 (slab limits), §5 (weak-graph passage), §7 (density, spectral projections), §8 (simple Perron root), §12 and §13 (covariance and parity via Lemma 2.8).
Dossier. `gj2-ground-existence-08`, `gjii-ground-12`, `gj71-lower-bound-02`, `gj71-theorem2-04`, `rosen-semigroup-22`, `rosen-selfadjoint-23`, `gjs-source-05/06` (E04–E06, E10–E13, E20).

Assumption A2 (Feynman–Kac–Nelson transfer)**.** Let $d\mu_0$ be the free Euclidean Gaussian measure with covariance $(-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$, J_t the sharp-time embeddings, and, for $h \in \mathcal{C}_{\mathrm{E}}$, $\mathcal{U}_h := \lambda \int h(z) {:}\mathscr{P}(\Phi(z)){:}\, dz$. For all bounded time-zero functions f, h', $T > 0$, and $g \in \mathcal{C}_{\mathrm{H}}$,

$$\langle f, e^{-TK(g)} h' \rangle = \int \overline{(J_T f)}\, (J_0 h')\, e^{-\mathcal{U}_{\mathbf{1}_{[0,T]} \otimes g}}\, d\mu_0,$$

with the interaction over exactly the slab $[0, T]$ carrying the two embeddings, together with the analogous multi-time formulas on finite slabs (used in §4 in the two- and one-time forms, where the symmetric slab cutoffs $h_{T,g} := \mathbf{1}_{[-T,T]} \otimes g \in \mathcal{C}_{\mathrm{E}}$ enter through the *normalized* measures $dq_{h_{T,g}}$).

Sources. Free-field FKN: [12, Thm. III.6]; interacting form: [12, Cor. V.14, Thm. V.15]; equivalently [10, Thms. II.14, II.16].

Scope. [12, Thm. III.12] is a projection-product estimate, *not* an FKN theorem, and is nowhere used (audit finding S1). The identification of sharp-time Wick functionals with Q-space multiplication operators is the conventions lock, Remark 2.5.

Used in. §4 (slab identities), §8 (path formula).
Dossier. `simon-free-fkn-111`, `simon-fkn-184/185`, `grs-fkn-39/40`, `simon-wrong-iii12-119` (E27–E43 range).

Assumption A3 (finite-volume stability)**.** For every $h \in \mathcal{C}_{\mathrm{E}}$: $\mathcal{U}_h \in L^p(d\mu_0)$ for every $p < \infty$, and $e^{-\varrho \mathcal{U}_h} \in L^1(d\mu_0)$ for every finite $\varrho > 0$. In particular $0 < Z(h) := \int e^{-\mathcal{U}_h} d\mu_0 < \infty$, and the interacting measures

$$dq_h := Z(h)^{-1} e^{-\mathcal{U}_h}\, d\mu_0, \qquad h \in \mathcal{C}_{\mathrm{E}},$$

are well-defined probability measures.

Source. [12, Thm. V.7] and its chapter context.

Scope. Only finite slabs and compactly supported h occur below; no infinite-volume stability is imported. A self-contained quantitative lower bound on $Z(h)$ and the h-uniform L^p bound on $e^{-\mathcal{U}_h}$ are re-proved in Appendix C.

Used in. §3 (Lemma 3.1), §4, §8 (Hölder/Vitali chains), Appendix C.
Dossier. `simon-exp-174`, `simon-transfer-178/179/180` (E33–E43 range).

Assumption A4 (free hypercontractivity)**.** The free time-zero Markov semigroup P_T satisfies

$$\|\mathsf{P}_T F\|_q \le \|F\|_p \qquad \text{whenever} \qquad \frac{q-1}{p-1} = e^{2m_0 T}, \quad 1 < p \le q < \infty.$$

More generally (the abstract form of the same source): for the second quantization $\Gamma(c)$ of a contraction c on the one-particle space of any Gaussian L^2-space, $\|\Gamma(c)F\|_q \le \|F\|_p$ whenever $\|c\|^2 \le (p-1)/(q-1)$; the display is the case $c = e^{-T\mu}$, and the Mehler case $c = e^{-t}\mathbf{1}$ on $L^2(d\mu_0)$ is the form used in Appendix C.

Sources. [12, Thm. I.17] (the abstract form and the display); the time-region form is [12, Thm. III.17].

Scope. [12, Thms. V.10, V.12] are, respectively, a transfer-matrix application and an essential-self-adjointness theorem; neither is the hypercontractivity input and neither is cited for it (audit finding S2).

Used in. §8 (the pairing bound), Appendix C (tail estimate).

Dossier. `simon-hyper-056`, `simon-timeslice-125/126`.

Assumption A5 (domain smoothing; common core)**.** *(i)* For every $g \in \mathcal{C}_{\mathrm{H}}$ and $t > 0$,

$$\operatorname{Ran} e^{-tK(g)} \subseteq D(H_0) \cap D(V')$$

for every polynomial multiplication operator V' of the time-zero field with coefficients in $L^1 \cap L^2$ of compact support (in particular for every $V(h)$ as in §2.1). *(ii)* For every affine path $g_s = (1-s)g + sg'$, $g, g' \in \mathcal{G}$, there is a common core $\mathscr{C}$ for all $K(g_s)$, $s \in [0,1]$, on which $K(g_s)\xi = K(g_{s_0})\xi + (s - s_0)V(g' - g)\xi$.

Sources. (i) is the model-specific smoothing theorem [6, Ch. 7, Thm. 7.4]. (ii) is part of the cutoff construction; essential self-adjointness pinpoints, in their roles per the audit: [12, Thm. V.12], [11, Thm. 5.3], [2].

Scope. [6, Thm. 7.3] alone assumes a potential bounded below and is therefore *not* sufficient here; the citation is to Theorem 7.4 (audit finding S3).

Used in. Proposition 2.6, §4 (dominators), §8 (eigenvalue differentiation).

Dossier. `gjexp-domain-165/166/167` (E17–E19).

Assumption A6 (free propagation; patching; the dynamics α)**.** *(i)* The free dynamics $\operatorname{Ad} e^{itH_0}$ has propagation speed one (Klein–Gordon finite propagation), and the Trotter product formula for $e^{itK(g)}$ converges strongly. *(ii)* There is a one-parameter automorphism group α of $\mathfrak{A}$, independent of g, with

$$\alpha_t(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|}), \qquad \alpha_t(A) = \beta_t^g(A) \ \text{ if } g \equiv 1 \text{ on } O_{|t|} \ (A \in \mathfrak{A}(O)). \tag{P}$$

Sources. [1, §III]; [2, §§3.5–3.6]; [6, §4.1 and Ch. 4].

Scope. (P) is the Glimm–Jaffe patching construction verbatim; the g-uniform propagation bound for the *cutoff* dynamics is not imported but derived (Proposition 2.7).

Used in. Proposition 2.7, §9 (locality identity), §10, §12.

Dossier. `gj1-locality-06`, `gjexp-054/055/056`, `gj2-cutoff-independence-38`, `gj2-local-algebra-40` (E03, E06, E14–E16).

Assumption A7 (covariance data)**.** Let U_a implement the space translation τ_a on Q-space and let $\Theta := \Gamma(-1)$, so that $(\Theta\psi)(q) = \psi(-q)$ in Q-space. Then, for every $g \in \mathcal{C}_{\mathrm{H}}$,

$$U_a K(g) U_a^* = K(\tau_a g), \qquad \Theta H_0 \Theta = H_0, \qquad \Theta V(g)\Theta = V(g) \ \text{ for even } \mathscr{P}.$$

Sources. [2, p. 398] (cutoff covariance); the transformation of Wick powers under $\Gamma(-1)$ is the parity of the polynomial.

Scope. Only the displayed operator identities are imported; the state-level consequences are derived in Lemma 2.8.

Used in. Lemma 2.8; thence §12, §13.

Dossier. `gj2-cutoff-independence-38` (E06).

Assumption A8 (the GJS package)**.** The Euclidean input is [7], used exactly as printed, as follows.

(i) Measures. For $h \in \mathcal{C}_{\mathrm{E}}$, $dq_h = Z(h)^{-1} e^{-\mathcal{U}_h} d\mu_0$ as in Assumption A3; the Hamiltonian cutoff class of [7, (1.6), p. 588] is $\mathcal{C}_{\mathrm{H}}$.

(ii) Observable class. The class (1.1.5) of [7] consists of sharp-time monomials $Q(x^0) = \int d\vec{x}\, w(\vec{x}) \prod_i {:}\Phi(x_i)^{m_i}{:}$ with L^2 kernels; equal times are allowed, and products of individually Wick-ordered powers are admissible without reordering. Equation (1.1.6) of [7], $Q = \int dx^0\, Q(x^0)$, defines the time-integrated monomials with kernels in L^2 of space-time; "localized" is defined for both forms by support in a product of unit lattice squares "centered at the lattice side [sic] $j_i \in \mathbb{Z}^{2}$" (p. 593), i.e. squares centred at integer points. The cells $[j, j+1)$ used in §§5–6 are this lattice shifted by one half unit in space; the count bounds used below (at most three cells for a support of length 2, Theorem 5.1, Step 2; $N_I \le \lfloor L \rfloor + 2$, Lemma 6.1) hold for any unit lattice, so the same computation on the squares of p. 593 gives the same constants. GJS bound the degree of each *individual* Wick factor: $n_i < \bar{n}$ for a fixed finite $\bar{n}$, any choice $\bar{n} \ge \deg \mathscr{P}$ being sufficient; the printed text adds "Since there is no restriction on r, Q still may have an arbitrarily large degree" (p. 593). Footnote 9 permits $\bar{n} = \infty$ at the price $N(\Delta)! \to (N(\Delta)!)^3$ in (1.1.7), (1.1.8), (2.4.5). Every observable used below has individual factor degrees $\le \max(2p', 1)$, so the fixed finite choice $\bar{n} = 2p' + 1$ applies and the printed first-power weights are used as displayed; the *total* degrees of the observables of §7 are unbounded, exactly as the printed sentence licenses.

(iii) Printed norms.

$$\begin{aligned} (1.1.7)\quad \|Q\|_{\mathrm{GJS}} &= K_1 \Big[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \Big] \|w\|_2, \\ (1.1.8)\quad \|Q(x_0)\|_{\mathrm{GJS}} &= K_1 \Big[\prod_{\Delta} K_2^{2N(\Delta)} N(\Delta)! \Big] \|w(x_0, \cdot)\|_2, \end{aligned} \tag{1}$$

where $N(\Delta)$ is the total degree localized in the unit square Δ, and (1.1.9): a nonlocalized polynomial is normed by summing its localized pieces. Replacing K_2 by $\max\{1, K_2\}$ only enlarges right-hand sides; *we fix* $K_2 \ge 1$.

(iv) CE2.

Source Fact 2.2 (CE2 = [7, Thm. 1.1.7], p. 594)**.** For localized monomials Q_1, Q_2 with disjoint localization squares separated by a strip of width d bounded by two parallel lines (any orientation),

$$\Big| \int Q_1 Q_2\, dq_h - \int Q_1 dq_h \int Q_2 dq_h \Big| \le C_7 e^{-m_1 d} \|Q_1\|_{\mathrm{GJS}} \|Q_2\|_{\mathrm{GJS}},$$

with $m_1 > 0$ and C_7 *independent of* h, Q_1, Q_2 (so stated in the source).

(v) CE1.

Source Fact 2.3 (CE1 = [7, Thm. 1.1.8], p. 595)**.** The printed estimate is

$$\Big| \int Q(x^0)\, dq_h \Big| \le C_8 \|Q(x^0)\|_{\mathrm{GJS}},$$

stated uniformly as the source cutoff tends to one. This item records the literal one-sentence statement of Theorem 1.1.8. The cutoff-uniform form consumed below is item (vi), read off the printed Theorem 1.1.11 (p. 596) in the specialization the source itself prints as its proof of Theorem 1.1.8 (p. 600).

(vi) The cutoff-uniform moment estimate U_{GJS} (GJS Theorem 1.1.11 at $B = I$, $n = 0$; imported as printed). For

$$\mathcal{C}_{\mathrm{E}}^{\mathrm{slab}} := \{h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g : T > 0,\ g \in \mathcal{C}_{\mathrm{H}}\} \subset \mathcal{C}_{\mathrm{E}},$$

one and the same constant C_8 satisfies

$$\left| \int Q(x^0)\, dq_{h_{T,g}} \right| \le C_8 \|Q(x^0)\|_{\mathrm{GJS}} \quad (T > 0,\ g \in \mathcal{C}_{\mathrm{H}}). \tag{U_{GJS}}$$

The quantifiers include every affine cutoff $g_s = (1-s)g + sg'$ used later, because $g_s \in \mathcal{C}_{\mathrm{H}}$. Equation ($U_{\mathrm{GJS}}$) is the CE1 uniformity needed in §§5 and 7; since 31 October 2026 it is read off one printed theorem of the source, as follows.

Source Fact ([7, pp. 596, 600 and 629])**.** *(a) p. 596.* "We let A and B denote monomials of type (1.1.5) of degree $|A|$ and $|B|$ and with kernels w_A and w_B respectively. We assume that A is localized at times $x_{i0} \le 0$, and that B is localized at times $x_{i0} \ge 0$. Let B_T denote the translate of B at time $T > 0$, so that B_T is localized at times $\ge T$." "In the next two results, we take all times $x_{1,0} = t = \cdots = x_{r,0}$ equal in $Q(x_0)$. To indicate this restriction, we write $Q(t)$ in place of $Q(x_0)$." [...] "THEOREM 1.1.11. Let n and $\varepsilon > 0$ be given and let $\gamma = (n+1)m_0(1-\varepsilon)$. For any A, we may choose n^{th} degree polynomials $Q_n(t_j)$ of the form (1.1.5) with $r = 1$, so that

$$\sum\nolimits_{t_j > 0} e^{\gamma t_j} \|Q_n(t_j)\| \le O(1)\|A\| \tag{1.1.14}$$

and so that

$$Q = \sum\nolimits_{0 < t_j < T} Q_n(t_j) \tag{1.1.15}$$

satisfies

$$\left| \int (A - Q) B_T\, dq_h \right| \le O(1)\|A\|\, \|B\|\, e^{-\gamma T}, \tag{1.1.16}$$

uniformly in h. We shall choose $t_j = j$, $j \in \mathbb{Z}$." *(b) p. 600.* "Proof of Theorem 1.1.8, assuming Theorem 1.1.10 [sic]. Choose $T = 0$ and $B = I$ in Theorem 1.1.11. Then $Q = 0$, so Theorem 1.1.8 is a special case of Theorem 1.1.10 [sic]." *(c) p. 629.* "Theorem 1.1.11 follows from this inequality, while Theorem 1.1.8 is the special case $B = I$, $n = 0$."

Reading (vi) off the source fact. Take $n = 0$ and $B = I$ in Theorem 1.1.11, as the source does on p. 600 ($B = I$, $T = 0$) and names on p. 629 ($B = I$, $n = 0$). For $0 < T < 1$ the sum (1.1.15) over the integers $0 < t_j < T$ is empty, so $Q = 0$ and $I_T = I$, and (1.1.16) reads $|\int A\, dq_h| \le O(1)\|A\|\, \|I\|\, e^{-\gamma T} \le O(1)\|A\|\, \|I\|$, uniformly in h, for every monomial A of type (1.1.5) localized at times ≤ 0; the source's own choice $T = 0$ (p. 600) gives the same bound, and $0 < T < 1$ is used here because the source introduces B_T for $T > 0$. By (1.1.7)–(1.1.8), $\|I\| = K_1$ (no Wick factors: the product is empty and the kernel is the constant 1), so (U_{GJS}) holds with

$$C_8 := O(1)\, K_1,$$

$O(1)$ the constant of Theorem 1.1.11 at $n = 0$ and a fixed $\varepsilon \in (0,1)$; it holds for every $h \in \mathcal{C}_{\mathrm{E}}$, in particular on $\mathcal{C}_{\mathrm{E}}^{\mathrm{slab}}$, and C_8 depends on the printed thresholds only. Three remarks

make the reading exact. *(1)* A sharp-time monomial at an arbitrary time x^0 is carried to a time in $(-1, 0]$ by an integer time translation, which maps the lattice squares of p. 593 onto themselves, preserves the norms (1.1.7)–(1.1.8), the class $\mathcal{C}_{\mathrm{E}}$ and the measure $d\mu_0$, and carries dq_h to $dq_{h'}$ with h' the translated cutoff; the bound at h' is the bound at h. *(2)* A general $Q(x^0)$ with kernel of bounded support is a finite sum of localized monomials, and $\|Q(x^0)\|_{\mathrm{GJS}}$ is the sum of their norms by (1.1.9); the bound is applied to each piece, whose times are all equal, as the source requires for Theorems 1.1.10–1.1.11 (p. 596). *(3)* The smallness hypotheses under which Theorem 1.1.11 holds are not printed in its statement; the source prints them with Theorem 4.1, from which Theorems 1.1.8 and 1.1.11 are proved on pp. 628–629 ("Let m_0, K_2 and K_1 be sufficiently large, and let λ be sufficiently small"), and on p. 600 ("convergence of our expansions requires both sufficiently small λ and sufficiently large m_0"); §2.6 fixes the working constants accordingly. The admissibility of the degree-zero $B = I$ rests on the source's own use of it (pp. 597, 600, 629) and on p. 593's "there is no restriction on r".

(vii) Printed corroboration of (vi).

Source Fact 2.4 ([7, p. 629, §4])**.** "It is important to work with a translation invariant Hamiltonian, in order that time translation preserve subspaces of bounded momentum. The estimates of Theorems 1.1.8, 1.1.11, 3.1 and 4.1 are uniform in the space cutoff h, and the path space integrals converge as $h \to 1$. Thus these theorems hold in the infinite volume limit, and we proceed to work in the limit $h = 1$."

(vii′) Printed use of the quantifier by the source (corroboration). The following passages show the source stating the cutoff-uniform quantifier of (U_{GJS}) and consuming it for arbitrary cutoff functions of its class, not only in the limit $h \to 1$. The source's $V(g)$ is the interaction $\mathcal{U}_g$ of item (i); its $\langle F\rangle_h$ is $\int F\, dq_h$.

Source Fact ([7, pp. 594–597 and 629])**.** *(a) Proof of Theorem 1.1.1, pp. 594–595.* "For test functions $f_i \in C_0^\infty(\mathcal{O})$ and for cutoff functions h_1 and h_2, we assert that

$$\Big|\int \prod_i \Phi(f_i)\, dq_{h_1} - \int \prod_i \Phi(f_i)\, dq_{h_2}\Big| \le O(1)e^{-m_1 d/2} \tag{1.1.11}$$

where $d = \operatorname{dist}(\mathcal{O}, \operatorname{suppt} h_1 - h_2)$, and then Theorem 1.1.1 follows. To prove (1.1.11) let $g = h_2 - h_1$, and define $g_i = g\chi_{\Delta_i}$, where χ_{Δ_i} is the characteristic function of Δ_i. Also we use the notation $\langle F\rangle_h = \int F\, dq_h$, and set $Q = \prod_i \Phi(f_i)$. Then

$$\begin{aligned}
\langle Q\rangle_{h_2} - \langle Q\rangle_{h_1} &= \int_0^1 d\alpha\, \frac{d}{d\alpha}\langle Q\rangle_{h_1+\alpha g} \\
&= -\int_0^1 d\alpha \big[\langle QV(g)\rangle_{h_1+\alpha g} - \langle Q\rangle_{h_1+\alpha g}\langle V(g)\rangle_{h_1+\alpha g}\big] \\
&= -\int_0^1 d\alpha \sum_{i\in\mathbb{Z}^2} \big[\langle QV(g_i)\rangle_{h_1+\alpha g} - \langle Q\rangle_{h_1+\alpha g}\langle V(g_i)\rangle_{h_1+\alpha g}\big].
\end{aligned} \tag{1.1.12}$$

We apply Theorem 1.1.7 to each term in the sum to obtain

$$|\langle Q\rangle_{h_2} - \langle Q\rangle_{h_1}| \le \sum_{i\in\mathbb{Z}^2\cap \operatorname{supp} g} O(1)e^{-m_1 \operatorname{dist}(\mathcal{O},\Delta_i)} \le O(1)e^{-m_1 d/2}.$$

As a special case, the cutoff Schwinger functions S_h converge as $h \to 1$."

(b) Proof of Theorem 1.1.7, pp. 596–597. "Proof of Theorem 1.1.7, assuming Theorems 1.1.8 and 1.1.11. Take $n = 0$ in Theorem 1.1.11 Then Q is a polynomial of degree zero, i. e., a constant. From the choice $B = I$, it follows that

$$Q = \int A\, dq_h + O(1)\|A\| e^{-m_0(1-\varepsilon)T}.$$

After rotating and translating the strip separating Q_1 and Q_2 in Theorem 1.1.7, and after choosing $A = Q_1$, $B_T = Q_2$ in Theorem 1.1.11, the proof follows from Theorem 1.1.8, (1.1.16) and the above determination of Q." The estimate (1.1.16), p. 596, reads "$|\int (A - Q) B_T\, dq_h| \le O(1)\|A\|\,\|B\| e^{-\gamma T}$, uniformly in h."

(c) p. 629. "Theorem 1.1.11 follows from this inequality, while Theorem 1.1.8 is the special case $B = I$, $n = 0$." The uniformity sentence of the same page is Source Fact 2.4.

(viii) Scaling. The unitary space-time dilation of [7, p. 590] sends (m_0, λ) to $(s^{-1}m_0, s^{-2}\lambda)$; the exact reduction of the dimensionless smallness hypothesis to the printed one is Proposition 3.2.

Scope. Every item of this assumption is a printed statement of [7], used at the scope at which it is printed; this is the standard under which Assumptions A1–A7 are imported, and no stricter one is available to this manuscript, since the printed proof of Theorem 1.1.7 itself obtains its h-independent constant from Theorem 1.1.8 at the same h (p. 597) while Theorem 1.1.8 is printed only "uniformly as $h \to 1$" (p. 595). Four points of scope are recorded. *(a) Item (vi).* From 16 August to 30 October 2026 the slab-uniform estimate (U_{GJS}) was carried as a named hypothesis of Theorem 12.2, documented on 5 September 2026 as the quantifier the source states on p. 629 and consumes on pp. 594–597 (items (vii), (vii′)) but "assembled from four pages of the source rather than read off one theorem". The source fact of item (vi) shows that it is read off one printed theorem, Theorem 1.1.11, whose estimate (1.1.16) is printed "uniformly in h", in the specialization the source itself prints as its proof of Theorem 1.1.8; the reading of 5 September 2026 used the same pages as evidence for Theorem 1.1.8 and is the same mathematics with different bookkeeping. Accordingly item (vi) is imported as printed, and Theorem 12.2 no longer carries it as a separate hypothesis. This is a re-sourcing, not a proof: the manuscript does not re-derive the cluster expansion, and Appendix C audits the source mechanism and proves two analytic prerequisites only. Items (vii) and (vii′) are kept as corroboration. *(b) Item (iv).* GJS Theorem 1.1.7 is printed for localized monomials separated by a strip of any orientation (p. 594). Its printed proof rotates the strip into the time direction (p. 597) and so reaches sharp-time monomials only across strips parallel to the time-zero line. Lemma 7.4 uses only that case. *(c)* The source's own proof of Theorem 1.1.7 obtains its constant from Theorem 1.1.8 at the same h (p. 597); this manuscript does not re-verify that proof, exactly as it re-verifies no proof behind Assumptions A1–A7. *(d) Three readings made explicit.* The quantifier "uniformly in h" of (1.1.16) is read over the class of p. 589 (compact support, $0 \le h \le 1$), the only class of space-time cutoffs h of the measures dq_h that the source defines, exactly as Theorem 1.1.7's "independent of h" is read; p. 629 states the uniformity of Theorems 1.1.8 and 1.1.11 "in the space cutoff h" (Source Fact 2.4). The constant $O(1)$ of (1.1.16) is read as depending on n, ε and the standing thresholds only, not on A, B, T or h, as the source's $O(1)$ is read throughout this manuscript (Source Facts 2.2–2.3); §7 needs this for observables of unbounded total degree. The stability input (2.4.34) of the source's proof (Appendix C, M6) enters Theorems 1.1.7 and 1.1.11 alike; under the import standard it belongs to the proofs this manuscript does not re-verify. The formerly illegible norm prefactor was settled by page-image inspection: K_1 enters to the first power and the local weight is $K_2^{2N(\Delta)}$ (gate R2, `m4/r2-check.tex`). All constants in this manuscript use the exact reading (1).

Used in. §3 (scaling), §4, §5 (CE1), §6 (norms), §7 (CE1, CE2), Appendix C.
Dossier. `gjs-source-05/06`, `gjs-p589-06`, `gjs-scaling-07`, `gjs-theorems-10/11/12`, `gjs-core-11/12`, `gjs-uniform-46`, `gjs-uniformity-46`, `gjs-proof-14` (E20–E26, E54); `gjs-thm1111-13` (p. 596), `gjs-proof118-17` (p. 600), `gjs-thm41-45` (p. 628), added 31 October 2026.

Remark 2.5 (conventions lock)**.** The sharp-time restriction of the Euclidean covariance equals C: $\int \frac{dk_0}{2\pi}(k_0^2+\mu(k)^2)^{-1} = (2\mu(k))^{-1}$ exactly. Hence sharp-time $d\mu_0$-Wick ordering coincides, constant by constant, with the C-Wick ordering used in $V(h)$, and sharp-time functionals of $\Phi(0,\cdot)$ are multiplication operators on $\mathcal{H}$. No Wick-mass conversion arises anywhere below.

2.5 Immediate consequences of the imports

Proposition 2.6 (domain smoothing at the ground state)**.** $\Omega_g \in D(V(h))$ *for every* $g \in \mathcal{C}_{\mathrm{H}}$ *and every real* $h \in L^1 \cap L^2$ *of compact support.*

Proof. By Assumption A5(i), for $t > 0$ the range of $e^{-tK(g)}$ is contained in $D(V(h))$. (Replacing $K(g)$ by $H(g) = K(g) - E(g)$ multiplies the semigroup by the nonzero scalar $e^{tE(g)}$ and does not change its range.) Since $K(g)\Omega_g = E(g)\Omega_g$,

$$\Omega_g = e^{tE(g)}e^{-tK(g)}\Omega_g \in \operatorname{Ran} e^{-tK(g)} \subseteq D(V(h)). \qquad \square$$

Proposition 2.7 (cutoff propagation, uniformly in g)**.** $\beta_t^g(\mathfrak{A}(O)) \subseteq \mathfrak{A}(O_{|t|})$ *for every* $g \in \mathcal{C}_{\mathrm{H}}$ *and* $t \in \mathbb{R}$*, with propagation speed* ≤ 1 *independent of* g*.*

Proof. The interaction automorphism $\operatorname{Ad} e^{itV(g)}$ has speed zero: for A generated by Weyl operators supported in $B \Subset O$ pick $\chi \in C_c^\infty(O)$, $\chi = 1$ near B; then $V(g) = V(\chi g) + V((1-\chi)g)$ with strongly commuting summands, $e^{itV(\chi g)} \in \mathfrak{A}(O)$, $e^{itV((1-\chi)g)} \in \mathfrak{A}(B)'$, so $\operatorname{Ad} e^{itV(g)}(A) \in \mathfrak{A}(O)$; extend by strong density. The free dynamics has speed one (Assumption A6(i)). The n-th Trotter approximant of $\beta_t^g(A)$ then lies in $\mathfrak{A}(O_{|t|})$, which is strongly closed, and the Trotter formula converges strongly (Assumption A6(i)). $\square$

Lemma 2.8 (covariance and parity of the cutoff states)**.** *For every* $g \in \mathcal{C}_{\mathrm{H}}$ *and* $a \in \mathbb{R}$*:*

$$\omega_{\tau_a g} = \omega_g \circ \tau_{-a}. \tag{T}$$

For even $\mathscr{P}$*:* $\omega_g \circ \theta = \omega_g$*.*

Proof. By Assumption A7, $U_a K(g) U_a^* = K(\tau_a g)$, so by simplicity (Assumption A1) $\Omega_{\tau_a g} = U_a\Omega_g$ up to a phase, which cancels in the state; this is (T).

For parity: $\Theta = \Gamma(-1)$ is unitary, self-adjoint, commutes with H_0, and for even $\mathscr{P}$ satisfies $\Theta V(g)\Theta = V(g)$ (Assumption A7); hence $\Theta\Omega_g = c\Omega_g$ with $c \in \{\pm 1\}$ by simplicity and $\Theta^2 = 1$. In Q-space $(\Theta\psi)(q) = \psi(-q)$, so $\Theta\Omega_g > 0$ a.e. together with $\Omega_g > 0$ a.e. (Assumption A1) forces $c = +1$. Thus $\omega_g(\theta(A)) = \langle \Theta\Omega_g, A\Theta\Omega_g\rangle = \omega_g(A)$. $\square$

2.6 Standing weak-coupling assumption

Fix source-admissible constants $M_* > 0$ and $\lambda_* > 0$ such that the hypotheses printed for the two theorems imported from [7] hold at bare mass M_* whenever $0 \leq \lambda' < \lambda_*$: for Theorem 1.1.7, "Let m_0, K_2 and K_1 be sufficiently large and let λ be sufficiently small" (p. 594); for Theorem 1.1.11, whose statement prints no smallness hypothesis, the same hypotheses,

printed with Theorem 4.1, from which it is proved (p. 628), and p. 600's "convergence of our expansions requires both sufficiently small λ and sufficiently large m_0". Define

$$\varepsilon_{\mathscr{P}} := \frac{\lambda_*}{M_*^2}.$$

Throughout, $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. By Proposition 3.2 and Remark 3.3 we may, and from now on do, assume that the pair (m_0, λ) itself satisfies the printed source hypotheses ($m_0 = M_*$ and $\lambda < \lambda_*$), so that the *unit-lattice* statements of Assumption A8 apply literally in the working coordinates; $m_1 > 0$ and C_7, C_8, K_1, K_2 denote the printed constants. Remark 3.3 records how each conclusion transports back to the original coordinates.

3 Dimensionless weak-coupling reduction

Lemma 3.1 (class compatibility)**.** (i) $\mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$. (ii) *For every $g \in \mathcal{C}_{\mathrm{H}}$ and every compact interval $I \subset \mathbb{R}$, the slab cutoff $\mathbf{1}_I \otimes g$ belongs to $\mathcal{C}_{\mathrm{E}}$; in particular $h_{T,g} = \mathbf{1}_{[-T,T]} \otimes g \in \mathcal{C}_{\mathrm{E}}$ and $\mathbf{1}_{[0,T]} \otimes g \in \mathcal{C}_{\mathrm{E}}$.* (iii) *For $g, g' \in \mathcal{C}_{\mathrm{H}}$ and $s \in [0,1]$, the affine path $g_s := (1-s)g + sg'$ stays in $\mathcal{C}_{\mathrm{H}}$, and $\mathbf{1}_I \otimes g_s = (1-s)\,\mathbf{1}_I \otimes g + s\,\mathbf{1}_I \otimes g'$ for every compact interval I, so that $s \mapsto \mathcal{U}_{\mathbf{1}_I \otimes g_s} = (1-s)\mathcal{U}_{\mathbf{1}_I \otimes g} + s\mathcal{U}_{\mathbf{1}_I \otimes g'}$ is affine. If moreover $g, g' \in \mathcal{G}$ and $\{g = 1\}^\circ \cap \{g' = 1\}^\circ \neq \varnothing$, then $g_s \in \mathcal{G}$ with $\{g_s = 1\}^\circ \supseteq \{g = 1\}^\circ \cap \{g' = 1\}^\circ$; in particular $d(g_s, O) \ge \min\{d(g, O), d(g', O)\}$ for every bounded interval O.*

Proof. (i) A function in $C_c^\infty(\mathbb{R})$ with values in $[0,1]$ is measurable with compact support. (ii) $\mathbf{1}_I \otimes g$ is a product of measurable functions, takes values in $[0,1]$, and $\operatorname{supp}(\mathbf{1}_I \otimes g) \subseteq I \times \operatorname{supp} g$ is compact. (iii) A convex combination of $[0,1]$-valued measurable functions is $[0,1]$-valued and measurable, and $\operatorname{supp} g_s \subseteq \operatorname{supp} g \cup \operatorname{supp} g'$ is compact; the identity $\mathbf{1}_I \otimes g_s = (1-s)\,\mathbf{1}_I \otimes g + s\,\mathbf{1}_I \otimes g'$ holds pointwise, and $h \mapsto \mathcal{U}_h$ is linear in h on the common $L^2(d\mu_0)$ domain guaranteed by Assumption A3. For the last statement: $g_s \in C_c^\infty$, $0 \le g_s \le 1$, and $\{g_s = 1\} \supseteq \{g = 1\} \cap \{g' = 1\}$ (for $s \in (0,1)$ a convex combination of two numbers in $[0,1]$ equals 1 if and only if both do, and for $s \in \{0,1\}$ the inclusion is trivial); passing to interiors gives the inclusion, which is a nonempty open set by hypothesis. The distance bound follows since $d(\cdot, O)$ is monotone in the plateau interior. □

Proposition 3.2 (dimensionless weak-coupling reduction)**.** *The standing inequality $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ implies the source smallness hypothesis of Assumption A8. More precisely, the GJS unitary dilation with $s = m_0/M_*$ sends*

$$(m_0, \lambda) \longmapsto (s^{-1}m_0, s^{-2}\lambda) = \left(M_*, \frac{M_*^2}{m_0^2}\lambda\right),$$

and the transformed coupling is strictly smaller than λ_*.

Proof. For $m > 0$, write the Euclidean covariance, as an identity of tempered distributions, as

$$C_m(z) = \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{e^{ip\cdot z}}{p^2 + m^2}\, dp.$$

Since $sM_* = m_0$, the substitution $q = sp$ gives, without an omitted power of s,

$$\begin{aligned} C_{M_*}(sz) &= \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{e^{ip\cdot sz}}{p^2 + M_*^2}\, dp \\ &= \frac{1}{(2\pi)^2} \int_{\mathbb{R}^2} \frac{s^{-2} e^{iq\cdot z}}{s^{-2}q^2 + M_*^2}\, dq = C_{m_0}(z). \end{aligned}$$

Thus the centered generalized Gaussian fields $\Phi_{m_0}(x)$ and $\Phi_{M_*}(sx)$ have identical joint covariances after smearing. This also fixes Wick ordering without a hidden constant. Indeed, for every regularized centered Gaussian variable X,

$$:e^{tX}: = \exp\bigl(tX - \tfrac{1}{2}t^2\mathbb{E}[X^2]\bigr)\,,$$

and the coefficient of t^n is $:X^n:/n!$. Apply the identity to a common mollifier, transported by $x \mapsto sx$, and then take the defining distributional limit. Equality of the regularized covariances shows that the unitary Gaussian identification sends $:\mathscr{P}(\Phi_{m_0}(x)):_{m_0}$ to $:\mathscr{P}(\Phi_{M_*}(sx)):_{M_*}$ with the coefficients of $\mathscr{P}$ unchanged. For a Euclidean cutoff h the change of variables $y = sx$ therefore gives

$$\lambda \int_{\mathbb{R}^2} h(x):\mathscr{P}(\Phi_{m_0}(x)):_{m_0}\,dx = s^{-2}\lambda \int_{\mathbb{R}^2} h(y/s):\mathscr{P}(\Phi_{M_*}(y)):_{M_*}\,dy.$$

The transformed cutoff $h_s(y) := h(y/s)$ still satisfies $0 \le h_s \le 1$ and has compact support, so $h_s \in \mathcal{C}_{\mathrm{E}}$. Hence the transformed parameters are exactly those displayed in the proposition, as stated in [7, p. 590]. Finally,

$$s^{-2}\lambda = \frac{M_*^2}{m_0^2}\lambda < M_*^2\varepsilon_{\mathscr{P}} = \lambda_*.$$

This is precisely the source smallness condition. If the source decay rate in the scaled coordinate $y = sx$ is $m_{1,*} > 0$, then $e^{-m_{1,*}d_y} = e^{-sm_{1,*}d_x}$; thus the rate in the original coordinates is $m_1 := sm_{1,*} > 0$. The transport of the remaining structure is Remark 3.3. □

Remark 3.3 (coordinate convention and transport of the conclusions)**.** The dilation behind Proposition 3.2 identifies the model at (m_0, λ) with the model at (M_*, λ'), $\lambda' = (M_*/m_0)^2\lambda < \lambda_*$, as follows. Let $s = m_0/M_*$, $y = sx$, and let σ_s be the scaling automorphism of the Weyl algebra, $\sigma_s(W(f,h)) = W(s^{-1}f(\cdot/s), h(\cdot/s))$ (a symplectic map). The sharp-time covariance is scale-covariant between the two masses, $C_{M_*}(s\,z) = C_{m_0}(z)$ (the one-dimensional analogue of the substitution in the proof of Proposition 3.2), so σ_s carries the mass-m_0 Fock state onto the mass-M_* Fock state and is unitarily implemented by some U_s with $U_s\Omega_0^{(m_0)} = \Omega_0^{(M_*)}$; moreover $\sigma_s(\mathfrak{A}(O)) = \mathfrak{A}(sO)$. On the generators, $\mu_{M_*}(k/s) = s^{-1}\mu_{m_0}(k)$ (since $sM_* = m_0$) gives $U_sH_0^{(m_0)}U_s^* = s\,H_0^{(M_*)}$, and the Euclidean Wick identification of Proposition 3.2, restricted to sharp time (Remark 2.5), gives $U_sV_{m_0,\lambda}(g)U_s^* = s\,V_{M_*,\lambda'}(g(\cdot/s))$; hence

$$U_s\,K_{m_0,\lambda}(g)\,U_s^* = s\,K_{M_*,\lambda'}(g(\cdot/s)), \qquad g \in \mathcal{C}_{\mathrm{H}}.$$

Consequently $g \mapsto g(\cdot/s)$ is an order isomorphism of $\mathcal{G}$ onto $\mathcal{G}$ (and a bijection of $\mathcal{C}_{\mathrm{H}}$), cutoff ground states correspond under U_s, and the two cutoff nets of states are carried onto each other. Every conclusion of Theorem 12.2, proved in the working (scaled) coordinates, therefore holds for the original model with the dictionary: localization $O \leftrightarrow sO$; plateau distance $d \leftrightarrow s\,d$, so the rate becomes $\Psi_{m_{1,*}}(s\,d)$, again decaying faster than every inverse power of d; dynamics $\alpha_t \leftrightarrow \alpha_{st}$, the light cone being form-invariant; gap $m_1 := s\,m_{1,*}$; and the local-moment clause of item (5) transports to supports of length $2/s$ — equivalently, it holds for supports of length 2 after splitting h by a partition of unity into at most $\lceil s\rceil + 1$ pieces of support length $\le 2/s$ (finite additivity of V), with constant $(\lceil s\rceil + 1)$ times the transported one. All displayed constants of §§4–10 and of Appendices A, C are the constants of the working coordinates, in which the unit-lattice geometry of Assumption A8 is literal.

4 Slab transfer

Throughout this section and §§5, 7, a cutoff $g \in \mathcal{C}_{\mathrm{H}}$ is fixed where limits $T \to \infty$ are taken; *no uniformity in g beyond the printed constants is used, and no gap is assumed.* Set

$$h_{T,g} := \mathbf{1}_{[-T,T]} \otimes g \in \mathcal{C}_{\mathrm{E}} \quad \text{(Lemma 3.1(ii))},$$
$$\psi_s := e^{-sH(g)}\Omega_0, \qquad \hat{\psi}_s := \psi_s/\|\psi_s\|, \qquad c_g := \langle \Omega_g, \Omega_0\rangle > 0$$

($c_g > 0$ by Assumption A1: $\Omega_g > 0$ a.e., $\Omega_0 \equiv 1$).

Lemma 4.1 (slab FKN; bounded nonnegative arguments)**.** *For $T > 0$, $t \in [0, 2T]$, $s \in [-T, T]$, and bounded nonnegative measurable functions F_1, F_2 of the time-zero field:*

$$\int F_1\big(\Phi(-\tfrac{t}{2}, \cdot)\big) F_2\big(\Phi(\tfrac{t}{2}, \cdot)\big) dq_{h_{T,g}} = \frac{\langle F_1 \psi_{T-t/2}, e^{-tH(g)} F_2 \psi_{T-t/2}\rangle}{\|\psi_T\|^2}, \tag{2}$$

$$\int F_1\big(\Phi(s, \cdot)\big)\, dq_{h_{T,g}} = \frac{\langle \psi_{T-s}, F_1\, \psi_{T+s}\rangle}{\|\psi_T\|^2}; \tag{3}$$

all energy shifts cancel in the ratios.

Proof. This is the Feynman–Kac–Nelson formula for $Z(h_{T,g})^{-1} e^{-\mathcal{U}_{h_{T,g}}} d\mu_0$ (Markov property of $d\mu_0$ along constant-time lines; Assumption A2; the measure is well-defined by Assumption A3 and Lemma 3.1(ii); the identification of sharp-time Wick functionals with Q-space multiplications is Remark 2.5). The unshifted formula contains $K(g)$ and the same factor $e^{-2TE(g)}$ in numerator and partition function; multiplying every semigroup by the corresponding $e^{aE(g)}$ replaces $K(g)$ by $H(g)$ and cancels that factor exactly. For bounded nonnegative F_i both sides are finite. □

Lemma 4.2 (signed bounded extension)**.** *Equations (2)–(3) hold for bounded signed measurable F_1, F_2.*

Proof. Write $F_i = F_i^+ - F_i^-$ and expand both sides into the same four (respectively two) finite nonnegative terms of Lemma 4.1; bilinearity (respectively linearity) proves the identities. □

Lemma 4.3 (unbounded extension)**.** (i) *For nonnegative measurable F_1 and F_2, the identities (2)–(3) hold in $[0, \infty]$.* (ii) *For real measurable F, define the clipped truncation $F^{[\ell]} := (-\ell) \vee (F \wedge \ell)$, $\ell > 0$. Whenever $F\psi_a \in \mathcal{H}$,*

$$\|(F - F^{[\ell]})\psi_a\|^2 = \int |F - F^{[\ell]}|^2 |\psi_a|^2\, d\phi_0 \longrightarrow 0 \qquad (\ell \uparrow \infty),$$

by dominated convergence with dominator $|F|^2|\psi_a|^2$; and on path space

$$|F_1(\Phi(s_1, \cdot))F_2(\Phi(s_2, \cdot))| \le \tfrac{1}{2}\big(F_1(\Phi(s_1, \cdot))^2 + F_2(\Phi(s_2, \cdot))^2\big)$$

is a dominated-convergence majorant whenever the two square moments are finite. Consequently the signed identities used below are limits of the bounded signed identities of Lemma 4.2, with no formal manipulation of products of unbounded operators.

Proof. (i) Apply Lemma 4.1 to $F_i \wedge \ell$ and let $\ell \uparrow \infty$; monotone convergence on both sides gives the identities in $[0, \infty]$. (ii) The Hilbert-space limit is dominated convergence as displayed. Membership $\psi_a \in D(F)$ is supplied at each application: finiteness of the corresponding $t = 0$ identity of part (i) over the a-slab, together with the slab bound (U_{GJS}), forces $F\psi_a \in \mathcal{H}$

(this is how it is verified in Theorem 5.1, Step 5, and in Lemma 7.1); for the interaction-type multiplications it also follows independently from Assumption A5(i). On path space, $F_i^{[\ell]}(\Phi(s_i,\cdot)) \to F_i(\Phi(s_i,\cdot))$ pointwise with the displayed majorant, whose integrability is exactly the finiteness of the two square moments; the cutoff-uniform estimate (U_{GJS}) supplies those square moments at every later application (§5, §7). Dominated convergence then passes Lemma 4.2 to the limit on both sides. □

Lemma 4.4 (slab vacuum; simplicity only)**.** *$\hat\psi_s \to \Omega_g$ in norm, $\|\psi_s\| \downarrow c_g$, and $\|\psi_{s'}\|/\|\psi_s\| \in [1, 1/c_g]$ for $0 \le s' \le s$; the ratio tends to 1 whenever $s, s' \to \infty$.* No isolation of the ground-state eigenvalue is used.

Proof. Set $P_g = |\Omega_g\rangle\langle\Omega_g|$. Since $P_g\Omega_0 = c_g\Omega_g$,

$$\psi_s = c_g\Omega_g + e^{-sH(g)}(1-P_g)\Omega_0,$$

and the two summands are orthogonal. If $\mu_g^\perp$ is the spectral measure of $(1-P_g)\Omega_0$, then

$$\|e^{-sH(g)}(1-P_g)\Omega_0\|^2 = \int_{(0,\infty)} e^{-2sE}\, d\mu_g^\perp(E) \longrightarrow 0$$

by dominated convergence. The measure has no atom at 0 by simplicity alone (Assumption A1); no positive distance from zero is used. It follows that $\psi_s \to c_g\Omega_g$, $\|\psi_s\| \to c_g$, and $\hat\psi_s \to \Omega_g$. The spectral formula also shows that $s \mapsto \|\psi_s\|$ is decreasing. Since $c_g \le \|\psi_s\| \le \|\psi_0\| = 1$, for $0 \le s' \le s$,

$$1 \le \frac{\|\psi_{s'}\|}{\|\psi_s\|} \le \frac{1}{c_g}.$$

If both indices tend to infinity, numerator and denominator tend to $c_g > 0$, so their ratio tends to 1. □

5 The uniform local interaction moment

Theorem 5.1 (uniform local interaction moment (M_2))**.** *Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$ and every real measurable h with $\|h\|_\infty \le 1$ and $\operatorname{supp} h$ contained in an interval of length 2,*

$$\|V(h)\Omega_g\| \le M, \qquad M^2 := 9\, C_8\, \mathfrak{K}(p')\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n|\Big)^2, \qquad \mathfrak{K}(p') := K_1 K_2^{8p'} (4p')!. \tag{M_2}$$

Proof. Step 1: the squared interaction as a GJS observable. In Euclidean notation define

$$Q_V := \lambda^2 \sum_{n,n'=0}^{2p'} a_n a_{n'} \iint h(x)h(y)\, {:}\Phi(0,x)^n{:}\,{:}\Phi(0,y)^{n'}{:}\, dx\, dy. \tag{4}$$

Q_V is exactly the multiplication function $V(h)^2$: expanding the square of $\lambda\sum_n a_n \int h(x){:}\Phi(0,x)^n{:}\, dx$ gives (4).

No Wick reordering of the product is performed or required; the observable class of Assumption A8(ii) permits products of individually Wick-ordered powers at equal times, and by Remark 2.5 the sharp-time Wick ordering coincides with the C-ordering of $V(h)$.

Step 2: cell count. The support of h meets at most three half-open unit spatial cells $[j, j+1)$ in positive measure. Indeed, if it met four distinct cells in positive measure, the

extreme indices would satisfy $j_{\max} \geq j_{\min}+3$, and the containing interval would have infimum $< j_{\min}+1$ and supremum $> j_{\max} \geq j_{\min}+3$, hence length > 2.

Step 3: local weight of one ordered cell pair. For one ordered pair of cells and degrees n, n', the GJS local weight of (1) is bounded by $K_1 K_2^{8p'}(4p')!$. Indeed the total local degree is $n+n' \leq 4p'$. If both variables belong to one square, the weight is $K_1 K_2^{2(n+n')}(n+n')!$; if to two squares, it is $K_1 K_2^{2n} n!\, K_2^{2n'} n'!$. In both cases

$$K_2^{2(n+n')} \leq K_2^{8p'} \quad (K_2 \geq 1), \qquad n!\, n'! \leq (n+n')! \leq (4p')!.$$

Step 4: the GJS norm of Q_V. Sum over the at most $3^2 = 9$ ordered cell pairs and all coefficient pairs. The kernel of the (n, n', Δ, Δ') piece is $w(x,y) = h(x)\mathbf{1}_\Delta(x)\, h(y)\mathbf{1}_{\Delta'}(y)$, and $\|w\|_2 = \|h\mathbf{1}_\Delta\|_2 \|h\mathbf{1}_{\Delta'}\|_2 \leq 1$ because $\|h\mathbf{1}_\Delta\|_2^2 = \int_\Delta h^2 \leq 1$ ($\|h\|_\infty \leq 1$, unit cell). Hence

$$\|Q_V\|_{\mathrm{GJS}} \leq 9\, K_1 K_2^{8p'} (4p')!\, \lambda^2 \Big(\sum_{n=0}^{2p'} |a_n| \Big)^2 = M^2 / C_8. \tag{5}$$

Step 5: transfer at time zero. For every $T > 0$, apply (2) at $t = 0$ with the two *individually nonnegative* functions $F_1 = F_2 = |V(h)|$ (Lemma 4.3(i)); their product is $V(h)^2 = Q_V$. Then (U_{GJS}) and (5) give

$$\|V(h)\hat{\psi}_T\|^2 = \int V(h)^2 \, dq_{h_{T,g}} \leq C_8 \|Q_V\|_{\mathrm{GJS}} \leq M^2. \tag{6}$$

In particular the left side is finite, so $\psi_T \in D(V(h))$.

Step 6: weak-graph passage to Ω_g. By (6), the family $(V(h)\hat{\psi}_T)_{T>0}$ is bounded by M; choose $T_k \to \infty$ with $V(h)\hat{\psi}_{T_k} \rightharpoonup y$ weakly. The graph of the closed operator $V(h)$ is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, hence weakly closed. Since $\hat{\psi}_{T_k} \to \Omega_g$ strongly (Lemma 4.4), the pair (Ω_g, y) lies in the graph; thus $\Omega_g \in D(V(h))$, $y = V(h)\Omega_g$, and weak lower semicontinuity of the norm gives

$$\|V(h)\Omega_g\| \leq \liminf_k \|V(h)\hat{\psi}_{T_k}\| \leq M. \qquad \square$$

Corollary 5.2 (φ^4 instance)**.** *For $\mathscr{P}(\xi) = \xi^4$ ($p' = 2$, $\sum_n |a_n| = 1$; degrees exactly $4+4=8$),*

$$M^2 = 9 \cdot 8!\, C_8 K_1 K_2^{16} \lambda^2 = 362\,880\, C_8 K_1 K_2^{16} \lambda^2.$$

6 GJS norms of sharp-time polynomial observables

Throughout this section, $Q = \prod_{i=1}^r \varphi_0(f_i)$ with real $f_i \in C_c^\infty(\mathbb{R})$, all supported in an interval I of length L, viewed at a common sharp time as an observable of the class of Assumption A8(ii). We call such Q *cylinder monomials* and their linear span, together with $\mathbf{1}$, *cylinder polynomials*. Set

$$N_I := \#\{j \in \mathbb{Z} : \ |I \cap [j, j+1)| > 0\}.$$

Lemma 6.1 (cell count)**.** *$N_I \leq \lfloor L \rfloor + 2 \leq L + 2$. The bound $L+1$ is false for nonintegral L: the interval $I = (-0.1, 0.1)$ has length 0.2 and meets both $[-1, 0)$ and $[0, 1)$ in positive measure.*

Proof. Let $j_{\min}$ and $j_{\max}$ be the smallest and largest indices of cells meeting I in positive measure, so $N_I \leq j_{\max} - j_{\min} + 1$. Positive-measure intersection with $[j_{\min}, j_{\min}+1)$ forces $\inf I < j_{\min}+1$, and with $[j_{\max}, j_{\max}+1)$ forces $\sup I > j_{\max}$; hence

$$L = \sup I - \inf I > j_{\max} - (j_{\min}+1) = j_{\max} - j_{\min} - 1.$$

The integer $j_{\max} - j_{\min} - 1$ is strictly less than L, hence at most $\lceil L \rceil - 1 \le \lfloor L \rfloor$. Consequently $N_I \le (j_{\max} - j_{\min} - 1) + 2 \le \lfloor L \rfloor + 2$. □

Proposition 6.2 (GJS norms of polynomial observables)**.**

$$\|Q\|_{\mathrm{GJS}} \le K_1 K_2^{2r} r!\, N_I^{r/2} \prod_{i=1}^{r} \|f_i\|_2, \tag{7}$$

$$\|Q^2\|_{\mathrm{GJS}} \le K_1 K_2^{4r} (2r)!\, N_I^{r} \prod_{i=1}^{r} \|f_i\|_2^2 =: \mathfrak{n}(Q)^2. \tag{8}$$

Proof. Expand the kernel of Q into cell assignments: decompose each $f_i = \sum_j f_i \mathbf{1}_{[j,j+1)}$, where only the N_I cells meeting I in positive measure contribute. For any fixed assignment of the r factors to cells, the localized piece has, by (1), the weight $K_1 \prod_\Delta K_2^{2N(\Delta)} N(\Delta)!$ with $\sum_\Delta N(\Delta) = r$; the product of local factorials is at most the factorial of the total degree, $\prod_\Delta N(\Delta)! \le r!$, and the K_2-exponents sum to $2r$. The localized kernel is the tensor product of the restrictions, with L^2 norm $\prod_i \|f_i \mathbf{1}_{[j_i,j_i+1)}\|_2$. Summing over assignments ((1.1.9) of Assumption A8(iii)) and using, for each i,

$$\sum_{j:\,|I\cap[j,j+1)|>0} \|f_i \mathbf{1}_{[j,j+1)}\|_2 \le N_I^{1/2} \|f_i\|_2$$

by Cauchy–Schwarz over the N_I cells, proves (7). Applying the same argument to the $2r$ factors of Q^2 (degree $2r$, K_2-exponent $4r$, factorial $(2r)!$, and $N_I^{1/2}$ per factor) proves (8). □

Remark 6.3 (row independence)**.** The norm of a sharp-time monomial is independent of the common sharp time s: at every s all factors occupy one temporal row, and changing s changes only that common row label, not the spatial cell assignments, local degrees, kernel norms, or the product of weights. This is the precise statement; arbitrary time translation does not literally preserve the fixed lattice of unit squares.

7 The uniform gap

Throughout this section $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, $g \in \mathcal{C}_{\mathrm{H}}$ is arbitrary and fixed, and $Q = \prod_{i=1}^r \varphi_0(f_i)$ is as in §6. Write $Q(s)$ for the same monomial at sharp Euclidean time s.

Lemma 7.1 (moment transfer)**.** $\sup_{T>0} \|Q\hat{\psi}_T\|^2 \le C_8\, \mathfrak{n}(Q)^2$*; moreover* $\Omega_g \in D(Q)$ *and* $\|Q\Omega_g\|^2 \le C_8\, \mathfrak{n}(Q)^2$.

Proof. Apply (2) at $t = 0$ with $F_1 = F_2 = |Q|$ (Lemma 4.3(i)); the Euclidean integrand is Q^2 at time 0, so (U_{GJS}), Remark 6.3 and (8) give

$$\|Q\hat{\psi}_T\|^2 = \int Q(0)^2\, dq_{h_{T,g}} \le C_8 \|Q^2\|_{\mathrm{GJS}} \le C_8\, \mathfrak{n}(Q)^2$$

for every T. Choose from the bounded family $Q\widehat{\psi}_T$ a weakly convergent subnet $Q\widehat{\psi}_{T_\alpha} \rightharpoonup y$. Since $\widehat{\psi}_{T_\alpha} \to \Omega_g$ strongly and the graph of the closed multiplication operator Q is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, hence weakly closed, (Ω_g, y) lies in that graph. Thus $\Omega_g \in D(Q)$, $y = Q\Omega_g$, and weak lower semicontinuity gives $\|Q\Omega_g\| \le \liminf_\alpha \|Q\widehat{\psi}_{T_\alpha}\| le C_8^{1/2} \mathfrak{n}(Q)$. □

Lemma 7.2 (one-point limit)**.** *For every fixed* $s \in \mathbb{R}$*,*

$$\int Q(s)\, dq_{h_{T,g}} \longrightarrow \langle \Omega_g, Q\Omega_g \rangle \qquad (T \to \infty).$$

Proof. For $T > |s|$, (3) (extended to Q by Lemma 4.3; the square moments $\int Q(s)^2\, dq_{h_{T,g}} \le C_8 \|Q^2\|_{\mathrm{GJS}}$ at the needed times follow from (U_{GJS}) with Remark 6.3, as in Lemma 7.1) gives

$$\int Q(s)\, dq_{h_{T,g}} = \Big\langle \hat{\psi}_{T-s},\, \frac{Q\psi_{T+s}}{\|\psi_T\|} \Big\rangle \frac{\|\psi_{T-s}\|}{\|\psi_T\|}. \tag{9}$$

The first vector tends strongly to Ω_g and the last factor tends to 1 (Lemma 4.4). Write the middle vector as

$$\frac{Q\psi_{T+s}}{\|\psi_T\|} = \frac{\|\psi_{T+s}\|}{\|\psi_T\|}\, Q\hat{\psi}_{T+s}. \tag{10}$$

If $s \ge 0$, the scalar ratio in (10) is at most 1; if $s < 0$, it is at most c_g^{-1}, both by Lemma 4.4. (The inequality $\|\psi_{T+s}\| \le \|\psi_T\|$ holds only for $s \ge 0$; the two signs must be treated separately.) Thus (10) is bounded in either case, by Lemma 7.1. Every weak subsequential limit of (10) equals $Q\Omega_g$, by weak closedness of the graph of Q, strong convergence $\hat{\psi}_{T+s} \to \Omega_g$, and because the scalar ratio tends to 1. Hence the whole middle vector converges weakly to $Q\Omega_g$, and (9) converges to $\langle \Omega_g, Q\Omega_g \rangle$. □

Lemma 7.3 (two-point lower semicontinuity)**.** *For every $t \ge 0$, with the Euclidean integrals over slabs of half-width $T \ge t/2$ (so that (2) applies),*

$$\langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle \le \liminf_{T\to\infty} \int Q(-t/2)\, Q(t/2)\, dq_{h_{T,g}}.$$

Proof. With $u_T := Q\psi_{T-t/2}/\|\psi_T\|$, the extension of (2) to Q (Lemma 4.3; square moments at the two times by (U_{GJS}) and Remark 6.3, as in Lemma 7.1) is the exact equality

$$\int Q(-t/2)\, Q(t/2)\, dq_{h_{T,g}} = \big\| e^{-tH(g)/2} u_T \big\|^2.$$

The ratio bound of Lemma 4.4 and Lemma 7.1 make $(u_T)_T$ bounded. Indeed,

$$u_T = \frac{\|\psi_{T-t/2}\|}{\|\psi_T\|}\, Q\widehat{\psi}_{T-t/2},$$

the scalar factor tends to 1 and is bounded by c_g^{-1}, and $\widehat{\psi}_{T-t/2} \to \Omega_g$ strongly. From any subnet of (u_T) choose a weakly convergent further subnet, say $u_{T_\alpha} \rightharpoonup u$. Along a further subnet the bounded vectors $Q\widehat{\psi}_{T_\alpha - t/2}$ converge weakly to the same u. Since the graph of the closed linear operator Q is a closed linear subspace of $\mathcal{H} \oplus \mathcal{H}$, it is weakly closed; hence (Ω_g, u) belongs to that graph and $u = Q\Omega_g$. Every weak cluster point is therefore $Q\Omega_g$, so the full bounded net satisfies $u_T \rightharpoonup Q\Omega_g$. Boundedness of $e^{-tH(g)/2}$ preserves the weak convergence $e^{-tH(g)/2} u_T \rightharpoonup e^{-tH(g)/2} Q\Omega_g$, and weak lower semicontinuity of the norm gives the claim. □

Lemma 7.4 (CE2 across a horizontal strip)**.** *For $t > 2$ and every $h \in \mathcal{C}_{\mathrm{E}}$,*

$$\Big| \int Q(-t/2) Q(t/2)\, dq_h - \int Q(-t/2)\, dq_h \int Q(t/2)\, dq_h \Big| \le C_7 e^{2m_1} e^{-m_1 t}\, \|Q\|_{\mathrm{GJS}}^2.$$

Proof. Use the finite localization decomposition prescribed by GJS (1.1.9):

$$Q = \sum_{a\in A} Q_a, \qquad \|Q\|_{\mathrm{GJS}} = \sum_{a\in A} \|Q_a\|_{\mathrm{GJS}},$$

where every Q_a is a localized monomial and A is finite because the kernels of the cylinder polynomial have compact spatial support and finite degree. By linearity of expectation,

$$\mathrm{Cov}_{dq_h}(Q(-t/2), Q(t/2)) = \sum_{a,b\in A} \mathrm{Cov}_{dq_h}(Q_a(-t/2), Q_b(t/2)).$$

The localization squares of $Q_a(-t/2)$ lie in the row of unit squares meeting $x^0 = -t/2$, and those of $Q_b(t/2)$ lie in the row meeting $x^0 = t/2$. The open strip $\{|x^0| < t/2 - 1\}$ has width $t-2$ and separates every such pair. Source Fact 2.2, applied separately to each (a, b), and the triangle inequality therefore give

$$\begin{aligned}
|\mathrm{Cov}_{dq_h}(Q(-t/2), Q(t/2))| &\le C_7 e^{-m_1(t-2)} \sum_{a,b\in A} \|Q_a(-t/2)\|_{\mathrm{GJS}} \|Q_b(t/2)\|_{\mathrm{GJS}} \\
&= C_7 e^{-m_1(t-2)} \left(\sum_a \|Q_a\|_{\mathrm{GJS}} \right)^2 \\
&= C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\mathrm{GJS}}^2,
\end{aligned}$$

where the second equality uses the row-independence proved in Remark 6.3. No application of the localized CE2 theorem to an unlocalized sum is made. □

Proposition 7.5 (semigroup decay of centered observables)**.** *Let $\psi_Q := Q\Omega_g - \langle \Omega_g, Q\Omega_g\rangle \Omega_g$. Then*

$$\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \le \widehat{C}_Q \, e^{-m_1 t} \qquad (t \ge 0), \qquad \widehat{C}_Q := \max\{ C_7 e^{2m_1} \|Q\|_{\mathrm{GJS}}^2, \; e^{3m_1} \|\psi_Q\|^2 \}.$$

Proof. Let $t > 2$. Since $e^{-tH(g)}\Omega_g = \Omega_g$ and $\langle \Omega_g, Q\Omega_g\rangle$ is real,

$$\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle = \langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle - \langle \Omega_g, Q\Omega_g \rangle^2.$$

Write $\int Q(-t/2)Q(t/2)\, dq_{h_{T,g}} = a_T + b_T$, where $b_T = \int Q(-t/2)\, dq_{h_{T,g}} \int Q(t/2)\, dq_{h_{T,g}}$ and a_T is the connected difference. Then, by Lemma 7.3 and $\liminf_T (a_T + b_T) \le \sup_T |a_T| + \lim_T b_T$,

$$\langle Q\Omega_g, e^{-tH(g)} Q\Omega_g \rangle \le \sup_T |a_T| + \lim_T b_T \le C_7 e^{2m_1} e^{-m_1 t} \|Q\|_{\mathrm{GJS}}^2 + \langle \Omega_g, Q\Omega_g \rangle^2,$$

using Lemma 7.4 for $\sup_T |a_T|$ and Lemma 7.2 at $s = \mp t/2$ fixed for $\lim_T b_T$. Subtracting the square proves the bound for $t > 2$. For $0 \le t \le 3$, positivity of $H(g)$ gives $\langle \psi_Q, e^{-tH(g)} \psi_Q \rangle \le \|\psi_Q\|^2 \le e^{3m_1} \|\psi_Q\|^2 e^{-m_1 t}$. The two cases cover $[0, \infty)$ and both bounds are $\le \widehat{C}_Q e^{-m_1 t}$. □

Lemma 7.6 (exponential moments)**.** *Let $d\mu_g := |\Omega_g|^2 \, d\phi_0$ and $X_f := \varphi_0(f)$ for real $f \in C_c^\infty(\mathbb{R})$, $f \ne 0$, supported in an interval of length L, and put $K' := K_2^2 N_I^{1/2} \|f\|_2 > 0$. Then*

$$\mathbb{E}_{\mu_g}\left[X_f^{2n} \right] \le C_8 K_1 \, (K')^{2n} (2n)! \qquad (n \ge 1),$$

and consequently

$$\mathbb{E}_{\mu_g} e^{s|X_f|} \le 2 \max\{1, C_8 K_1\} \sum_{n=0}^{\infty} (sK')^{2n} < \infty \qquad (0 \le s < 1/K').$$

The factor $C_8 K_1$ affects only the prefactor, not the radius $1/K'$.

Proof. $\mathbb{E}_{\mu_g}[X_f^{2n}] = \|X_f^n \Omega_g\|^2 \le C_8\, \mathfrak{n}(X_f^n)^2$ by Lemma 7.1 applied to $Q = X_f^n$; and by (8) with $r = n$ and all $f_i = f$,

$$\mathfrak{n}(X_f^n)^2 = K_1 K_2^{4n} (2n)!\, N_I^n \|f\|_2^{2n} = K_1 (K')^{2n} (2n)!.$$

For the exponential bound: $e^{s|x|} \le e^{sx} + e^{-sx} = 2\cosh(sx)$, and

$$\mathbb{E}_{\mu_g} \cosh(sX_f) = \sum_{n\ge 0} \frac{s^{2n}}{(2n)!} \mathbb{E}_{\mu_g}[X_f^{2n}] \le \max\{1, C_8 K_1\} \sum_{n\ge 0} (sK')^{2n},$$

where the maximum covers the $n = 0$ term $\mathbb{E}_{\mu_g}[1] = 1$. The series converges for $sK' < 1$. □

Theorem 7.7 (density of polynomial vectors)**.** *Cylinder polynomials are dense in $L^2(d\mu_g)$. Equivalently, since $\Omega_g > 0$ a.e. makes multiplication by Ω_g a unitary from $L^2(d\mu_g)$ onto $\mathcal{H}$, the linear span of Ω_g and the vectors $Q\Omega_g$ (Q a cylinder monomial) is dense in $\mathcal{H}$.*

Proof. Step 1: vanishing conditional expectations. Let $\zeta \in L^2(d\mu_g)$ be orthogonal to all cylinder polynomials. Fix real $e_1, \dots, e_k \in C_c^\infty(\mathbb{R})$ and define

$$F(z_1, \dots, z_k) := \mathbb{E}_{\mu_g}\Big[\overline{\zeta}\, \exp\Big(\sum_{j=1}^k z_j X_{e_j}\Big)\Big].$$

By Cauchy–Schwarz, Hölder, and Lemma 7.6, F is defined and analytic in a nonempty polystrip $\{|\mathrm{Re}\, z_j| < s_0\}$ about $i\mathbb{R}^k$; differentiation under the integral is justified by the exponential dominator of Lemma 7.6. Every derivative of F at zero is an inner product of ζ with a cylinder polynomial, hence zero. Repeated application of the one-variable identity theorem gives $F = 0$ throughout the connected polystrip, in particular on $i\mathbb{R}^k$. Thus the finite complex measure

$$B \longmapsto \mathbb{E}_{\mu_g}\big[\overline{\zeta}\, \mathbf{1}_{\{(X_{e_1}, \dots, X_{e_k}) \in B\}}\big]$$

on $\mathbb{R}^k$ has vanishing Fourier transform, so it is zero by uniqueness of Fourier transforms of finite measures; equivalently $\mathbb{E}_{\mu_g}[\zeta \mid \sigma(X_{e_1}, \dots, X_{e_k})] = 0$.

Step 2: exhaustion. Choose $(e_j)_{j\ge 1} \subseteq C_c^\infty(\mathbb{R})$, real and dense in real $L^2(\mathbb{R})$ (C_c^∞ is dense there). The time-zero covariance $C = (2\mu)^{-1}$ is bounded, so

$$\mathbb{E}_{\phi_0}|X_f - X_e|^2 = \langle f - e, C(f - e)\rangle \le \|C\|\, \|f - e\|_2^2.$$

For an approximating sequence $e_{j_l} \to f$ in L^2, a subsequence of $X_{e_{j_l}}$ converges ϕ_0-almost surely, hence μ_g-almost surely because $\mu_g \ll \phi_0$. Therefore every X_f is measurable with respect to $\bigvee_k \sigma(X_{e_1}, \dots, X_{e_k})$ modulo null sets, and this sigma-algebra exhausts the cylinder sigma-algebra.

Step 3: martingale convergence. The conditional expectations of Step 1 vanish for every k and converge in $L^2(d\mu_g)$ to the conditional expectation on the exhausted sigma-algebra, which is ζ itself. Hence $\zeta = 0$, proving density. □

Theorem 7.8 (uniform gap (G))**.** *Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. For every $g \in \mathcal{C}_{\mathrm{H}}$,*

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [m_1, \infty), \tag{G}$$

and 0 is a simple eigenvalue.

Proof. Fix $0 \le \beta < m_1$ and let $P_\beta := \mathbf{1}_{[0,\beta]}(H(g))$.

Step 1: $P_\beta \psi_Q = 0$ for every centered polynomial vector. By the spectral theorem and Proposition 7.5, for every $t \ge 0$,

$$e^{-\beta t}\|P_\beta \psi_Q\|^2 \le \langle \psi_Q, e^{-tH(g)}\psi_Q\rangle \le \widehat{C}_Q\, e^{-m_1 t}.$$

Multiply by $e^{\beta t}$ and let $t \to \infty$; since $m_1 - \beta > 0$, $\|P_\beta \psi_Q\| = 0$. (The restriction $\beta \ge 0$ is necessary: for $\beta < 0$, $P_{[0,\beta]} = 0$, and Step 2 below fails.)

Step 2: $P_\beta \Omega_g = \Omega_g$, since $H(g)\Omega_g = 0$ and $0 \in [0, \beta]$.

Step 3: $P_\beta = |\Omega_g\rangle\langle\Omega_g|$. By Theorem 7.7, the complex span of Ω_g and the centered vectors ψ_Q is dense in $\mathcal{H}$. Steps 1–2 show that P_β maps this dense subspace into $\mathbb{C}\Omega_g$; continuity of the bounded projection extends the containment to $\mathcal{H}$, and $P_\beta \Omega_g = \Omega_g$ shows the range is exactly $\mathbb{C}\Omega_g$.

Step 4: no spectrum in $(0, m_1)$. If an open interval $J \Subset (0, m_1)$ carried a nonzero spectral projection, choose $\beta < m_1$ above the right endpoint of J; then $\mathbf{1}_J(H(g)) \le P_\beta - |\Omega_g\rangle\langle\Omega_g| = 0$, a contradiction. Hence (G) holds; simplicity of 0 is Assumption A1. □

8 Affine-path regularity

Throughout §§8–10, $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, and $\gamma > 0$ denotes a constant for which

$$\operatorname{spec} H(g) \subseteq \{0\} \cup [\gamma, \infty), \qquad \dim\ker H(g) = 1, \qquad \text{for every } g \in \mathcal{C}_{\mathrm{H}}; \tag{G_γ}$$

by Theorem 7.8 this holds with $\gamma = m_1$, and γ is instantiated there in §11. Fix $g, g' \in \mathcal{G}$ equal to 1 on a common nonempty open interval, and put

$$g_s := (1-s)g + sg', \quad v := g' - g, \quad K_s := K(g_s), \quad E_s := E(g_s), \quad H_s := K_s - E_s,$$

$$\Omega_s := \Omega_{g_s}, \qquad P_s := |\Omega_s\rangle\langle\Omega_s|, \qquad Q_s := 1 - P_s.$$

By Lemma 3.1(iii), $g_s \in \mathcal{G} \subset \mathcal{C}_{\mathrm{H}}$ for every $s \in [0,1]$, so all results of §§4–7 apply to every K_s.

Proposition 8.1 (path regularity of the semigroup)**.** *For each $T > 0$, the family $s \mapsto e^{-TK_s}$ is norm-continuous on $[0,1]$ and norm-C^1 on $(0,1)$, with derivative given by the FKN kernel of $-\mathcal{U}_v w_s$ in the notation of the proof.*

Proof. On free Euclidean path space over the slab $[0,T]$ define

$$\mathcal{U}_s := \lambda \int_0^T \int_{\mathbb{R}} g_s(x) :\mathscr{P}(\Phi(t,x)): dx\, dt, \qquad w_s := e^{-\mathcal{U}_s}, \qquad \mathcal{U}_v := \mathcal{U}_1 - \mathcal{U}_0,$$

so that $s \mapsto \mathcal{U}_s = (1-s)\mathcal{U}_0 + s\mathcal{U}_1$ is affine (Lemma 3.1(iii)). For bounded time-zero f, h, the interacting FKN formula of Assumption A2 reads

$$\langle f, e^{-TK_s} h\rangle = \int \overline{(J_T f)}\, (J_0 h)\, w_s\, d\mu_0, \tag{11}$$

with J_t the sharp-time embeddings.

Step 1: the pairing bound. Set $a := e^{-m_0 T}$ and

$$q_T := \frac{2}{1-a}, \quad r_T := \frac{2}{1+a}, \quad p_T := 1 + a, \quad p_T' := 1 + a^{-1}.$$

Then $1/r_T + 1/q_T = 1$, $r_T p_T = 2$, and $(p_T' - 1)/(p_T - 1) = a^{-2} = e^{2m_0 T}$, so Assumption A4 gives $\|\mathsf{P}_T F\|_{p_T'} \le \|F\|_{p_T}$ for the free Markov semigroup P_T on $L^2(d\phi_0)$. The Markov property of $d\mu_0$ along constant time lines and Hölder's inequality (note $1/p_T + 1/p_T' = 1$) yield

$$\|(J_T f)(J_0 h)\|_{r_T}^{r_T} = \int |h|^{r_T}\, \mathsf{P}_T\big(|f|^{r_T}\big)\, d\phi_0 \le \big\||h|^{r_T}\big\|_{p_T} \big\|\mathsf{P}_T(|f|^{r_T})\big\|_{p_T'} \le \|h\|_2^{r_T}\, \|f\|_2^{r_T},$$

using $\||h|^{r_T}\|_{p_T} = \|h\|_{r_T p_T}^{r_T} = \|h\|_2^{r_T}$ and hypercontractivity on the second factor. Hence

$$\|(J_T f)(J_0 h)\|_{r_T} \le \|f\|_2 \|h\|_2. \tag{12}$$

Step 2: L^a control and continuity of $s \mapsto w_s$. Finite-volume stability (Assumption A3, applicable since the slab cutoffs lie in $\mathcal{C}_{\mathrm{E}}$ by Lemma 3.1(ii)) gives $\mathcal{U}_v \in L^p(d\mu_0)$ for every finite p and $e^{-\varrho \mathcal{U}_i} \in L^1(d\mu_0)$ for every finite $\varrho > 0$, $i = 0, 1$. Since $\mathcal{U}_s$ is affine, Hölder's inequality with exponents $\frac{1}{1-s}, \frac{1}{s}$ gives

$$\|w_s\|_\varrho^\varrho \le \Big(\int e^{-\varrho \mathcal{U}_0} d\mu_0\Big)^{1-s} \Big(\int e^{-\varrho \mathcal{U}_1} d\mu_0\Big)^{s}, \tag{13}$$

a bound uniform in $s \in [0,1]$ for each ϱ. If $s_n \to s$, then $\mathcal{U}_{s_n} \to \mathcal{U}_s$ in probability (indeed in L^2), so $w_{s_n} \to w_s$ in probability; together with (13) at a larger exponent, Vitali's convergence theorem gives $w_{s_n} \to w_s$ in L^ϱ for every finite ϱ.

Step 3: the difference quotient. The calculus identity

$$\frac{w_{s+\delta} - w_s}{\delta} = -\mathcal{U}_v \int_0^1 w_{s+\vartheta\delta}\, d\vartheta \tag{14}$$

holds pointwise on path space. Choose finite p, ϱ with $q_T^{-1} = p^{-1} + \varrho^{-1}$. By Hölder, the L^{q_T} distance of (14) from $-\mathcal{U}_v w_s$ is at most $\|\mathcal{U}_v\|_p \int_0^1 \|w_{s+\vartheta\delta} - w_s\|_\varrho\, d\vartheta$, which tends to 0 as $\delta \to 0$ for $0 < s < 1$ by Step 2 (dominated convergence in ϑ, using (13)). Thus $s \mapsto w_s$ is L^{q_T}-differentiable on $(0,1)$ with derivative $-\mathcal{U}_v w_s$, and L^{q_T}-continuous on $[0,1]$.

Step 4: duality. By (11), (12), and Hölder with the conjugate pair (r_T, q_T),

$$\big|\langle f, (e^{-TK_s} - e^{-TK_r})h\rangle\big| \le \|(J_T f)(J_0 h)\|_{r_T}\, \|w_s - w_r\|_{q_T} \le \|f\|_2 \|h\|_2\, \|w_s - w_r\|_{q_T},$$

so $\|e^{-TK_s} - e^{-TK_r}\| \le \|w_s - w_r\|_{q_T}$, and the same estimate applied to the difference quotients transfers the L^{q_T} regularity of Step 3 to the operator norm. This proves the asserted norm continuity and differentiability. □

Proposition 8.2 (projection and eigenvector regularity; derivative formulas)**.** (i) *$s \mapsto P_s$ is norm-continuous on $[0,1]$ and norm-C^1 on $(0,1)$.* (ii) *Locally on $(0,1)$ one may choose normalized C^1 vectors Ω_s in the parallel gauge $\langle \Omega_s, \dot\Omega_s\rangle = 0$, and then*

$$\dot E_s = \langle \Omega_s, V(v)\Omega_s\rangle, \qquad \dot\Omega_s = -H_s^{-1} Q_s V(v)\Omega_s, \tag{15}$$

with $\|H_s^{-1} Q_s\| \le \gamma^{-1}$.

Proof. (i) Let $L_s := e^{-TK_s}$ for a fixed $T > 0$. By Assumption A1, L_s has the simple largest eigenvalue $\lambda_s = e^{-TE_s} = \|L_s\|$, and by ($\mathrm{G}_\gamma$),

$$\operatorname{spec} L_s \setminus \{\lambda_s\} \subseteq [0, e^{-T\gamma}\lambda_s].$$

The relative spectral separation $1 - e^{-T\gamma}$ is uniform in s. Since $s \mapsto L_s$ is norm-continuous, so is $s \mapsto \|L_s\| = \lambda_s$; a Riesz contour of relative radius independent of s around λ_s therefore

represents P_s as a norm-convergent Cauchy integral of the resolvent of L_s, locally in s, and the norm regularity of L_s (Proposition 8.1) passes to P_s through the contour integral. This gives exactly the stated regularity of P_s.

(ii) Let $\mathscr{C}$ be the common core of Assumption A5(ii). For $\xi \in \mathscr{C}$, $K_s\xi = K_{s_0}\xi + (s - s_0)V(v)\xi$. Differentiating the weak eigenvalue identity $\langle K_s\xi, \Omega_s\rangle = E_s\langle \xi, \Omega_s\rangle$ at $s = s_0$ (all terms differentiable: Ω_s is C^1 by (i) and the gauge; and $E_s = -T^{-1}\log\lambda_s$ is C^1 because $\lambda_s = \langle \Omega_s, L_s\Omega_s\rangle > 0$ is a composition of the C^1 maps of (i) and Proposition 8.1) gives

$$\langle H_s\xi, \dot\Omega_s\rangle = \langle \xi, (\dot E_s - V(v))\Omega_s\rangle, \tag{16}$$

where $\Omega_s \in D(V(v))$ by Proposition 2.6, so the right side is well-defined. Since $\mathscr{C}$ is a core for the self-adjoint H_s, (16) says $\dot\Omega_s \in D(H_s)$ and $H_s\dot\Omega_s = (\dot E_s - V(v))\Omega_s$. Pairing with Ω_s and using $H_s\Omega_s = 0$, $\langle \Omega_s, \dot\Omega_s\rangle = 0$ proves the first formula in (15). The parallel gauge puts $\dot\Omega_s \in Q_s\mathcal{H}$, and ($G_\gamma$) makes $H_s^{-1}Q_s$ bounded by γ^{-1}; applying $H_s^{-1}Q_s$ to $H_s\dot\Omega_s = (\dot E_s - V(v))\Omega_s$ and using $Q_s\Omega_s = 0$ proves the second formula. □

9 Filter, spectral separation, and clustering

Lemma 9.1 (exact inverse-energy filter)**.** *Choose an even $\chi \in C^\infty(\mathbb{R}; [0,1])$ with $\chi = 0$ on $[-\gamma/2, \gamma/2]$ and $\chi = 1$ outside $(-\gamma, \gamma)$, and set*

$$\widehat{W}_\gamma(E) := \begin{cases} \chi(E)/E, & E \neq 0, \\ 0, & E = 0, \end{cases} \qquad W_\gamma(t) := \frac{1}{2\pi}\int_{\mathbb{R}} e^{-itE}\,\widehat{W}_\gamma(E)\,dE. \tag{17}$$

Then: (i) $W_\gamma \in L^1(\mathbb{R})$ *and, for every $q \geq 0$, $\int_{\mathbb{R}}(1+|t|)^q|W_\gamma(t)|\,dt < \infty$; in particular*

$$T_\gamma(r) := \int_{|t|>r} |W_\gamma(t)|\,dt \leq C_{q,\gamma}(1+r)^{-q} \qquad (q \geq 0,\ r \geq 0).$$

(ii) *If H is self-adjoint with* $\operatorname{spec} H \subseteq \{0\} \cup [\gamma, \infty)$*, then*

$$H^{-1}(1 - P_{\ker H}) = \int_{\mathbb{R}} W_\gamma(t)\, e^{itH}(1 - P_{\ker H})\,dt, \tag{18}$$

a norm-convergent Bochner integral.

Proof. Throughout, $\widehat{W}_\gamma \in L^2(\mathbb{R})$ (it vanishes on $[-\gamma/2, \gamma/2]$, is bounded by $2/\gamma$, and decays like $|E|^{-1}$), and (17) is understood as its L^2-inverse Fourier transform; the symmetric improper integrals below converge (Dirichlet's test) and represent it almost everywhere, which justifies the manipulations.

(i) Oddness of $\widehat{W}_\gamma$ gives

$$W_\gamma(t) = -\frac{i}{\pi}\int_0^\infty \frac{\sin(tE)\,\chi(E)}{E}\,dE.$$

For $0 < |t| \leq 1$: split at $E = \gamma$. The transition part $\int_{\gamma/2}^{\gamma} |\sin(tE)|\chi(E)E^{-1}dE$ is absolutely bounded by a constant depending only on γ; for the tail $\int_\gamma^\infty \sin(tE)E^{-1}dE$, substitute $u = |t|E$ and use the uniform boundedness of $\int_c^d u^{-1}\sin u\,du$ over $0 < c < d$ (Dirichlet's test). Hence W_γ is bounded near $t = 0$, so integrable there. For $|t| \geq 1$: every derivative $\widehat{W}_\gamma^{(k)}$ with $k \geq 1$ is integrable on $\mathbb{R}$ (it is smooth, supported off $(-\gamma/2, \gamma/2)$, and decays like $|E|^{-1-k}$), so k integrations by parts in (17) give

$$|W_\gamma(t)| \leq \frac{\|\widehat{W}_\gamma^{(k)}\|_{L^1}}{2\pi|t|^k}.$$

Choosing $k > q+1$ proves the weighted integrability and, by Markov's inequality applied to the weight $(1+|t|)^q$, the tail bound for T_γ.

(ii) By construction $\widehat{W}_\gamma(E) = E^{-1}$ on $[\gamma,\infty)$ and $\widehat{W}_\gamma(0) = 0$. Fourier inversion and the spectral theorem give (18) on the spectral subspace of $[\gamma,\infty)$ and on $\ker H$ separately; the integral converges in norm because $W_\gamma \in L^1$ and $\|e^{itH}(1 - P_{\ker H})\| \le 1$. □

Lemma 9.2 (spectral separation)**.** *Let*

$$F(t) = \int_{[\gamma,\infty)} e^{itE}\, d\rho(E), \qquad G(t) = \int_{(-\infty,-\gamma]} e^{itE}\, d\sigma(E),$$

with complex measures obeying $|\rho|(\mathbb{R}), |\sigma|(\mathbb{R}) \le \mathfrak{m}$*, and suppose* $F = G$ *on* $(-R,R)$ *with* $\gamma R \ge 1$*. Then*

$$|F(0)| \le 2\,\mathfrak{m}\, e^{-\gamma R/2}. \tag{19}$$

Proof. Step 1: the Gaussian separator. For $a, \varepsilon > 0$ define

$$f_{a,\varepsilon}(t) := \frac{1}{2\pi i}\, \frac{e^{-t^2/(2a)}}{t - i\varepsilon}.$$

The transform of $(2\pi i)^{-1}(t - i\varepsilon)^{-1}$ is $\mathbf{1}_{(0,\infty)}(E)\, e^{-\varepsilon E}$ (one-pole contour integral); multiplication by the Gaussian in t convolves this with the normalized Gaussian, so

$$\check{f}_{a,\varepsilon}(E) = \int_0^\infty \sqrt{\frac{a}{2\pi}}\, e^{-a(E-E')^2/2}\, e^{-\varepsilon E'}\, dE' \in [0,1], \tag{20}$$

which tends to the standard normal distribution function $\Phi(\sqrt{a}E)$ as $\varepsilon \downarrow 0$.

Step 2: the tail of the separator. For $t \ge R$, $|t - i\varepsilon| \ge t$ and $t^{-1} \le t/R^2$, so

$$\int_{|t|\ge R} |f_{a,\varepsilon}(t)|\, dt \le \frac{1}{\pi R^2} \int_R^\infty t e^{-t^2/(2a)}\, dt = \frac{a}{\pi R^2}\, e^{-R^2/(2a)}. \tag{21}$$

Step 3: exchange of test function. Pair $f_{a,\varepsilon}$ with $F - G$. By hypothesis the pairing is supported in $|t| \ge R$; Fubini (justified by the finite total variations and $f_{a,\varepsilon} \in L^1$) and then $\varepsilon \downarrow 0$ with dominated convergence in energy give

$$\Big| \int \Phi(\sqrt{a}E)\, d\rho(E) - \int \Phi(\sqrt{a}E)\, d\sigma(E) \Big| \le \frac{2\,\mathfrak{m}\, a}{\pi R^2}\, e^{-R^2/(2a)}. \tag{22}$$

Step 4: optimization. On the support of ρ, $E \ge \gamma$, so $1 - \Phi(\sqrt{a}E) \le \frac{1}{2} e^{-a\gamma^2/2}$; on the support of σ, $E \le -\gamma$, so $\Phi(\sqrt{a}E) \le \frac{1}{2} e^{-a\gamma^2/2}$. Inserting and subtracting $\int d\rho = F(0)$ in (22),

$$|F(0)| \le \mathfrak{m}\, e^{-a\gamma^2/2} + \frac{2\,\mathfrak{m}\, a}{\pi R^2}\, e^{-R^2/(2a)}.$$

Set $a = R/\gamma$: both exponentials become $e^{-\gamma R/2}$, and $2a/(\pi R^2) = 2/(\pi\gamma R) \le 2/\pi < 1$ by $\gamma R \ge 1$; the two terms sum to at most $2\,\mathfrak{m}\, e^{-\gamma R/2}$, proving (19). □

Lemma 9.3 (clustering with one unbounded local argument)**.** *Let* $g \in \mathcal{C}_{\mathrm{H}}$*, write* $H = H(g)$*,* $\Omega = \Omega_g$*,* $\beta_t = \beta_t^g$*, and assume* (G_γ)*. Let* $B \subset \mathbb{R}$ *be compact, let* $X = X^*$ *be affiliated with* $\mathfrak{A}(U)$ *for every bounded open* $U \supseteq B$*, with* $\Omega \in D(X)$*, and let* $A \in \mathfrak{A}(O)$ *for a bounded interval* O*. If* $R := \operatorname{dist}(B, O) > 0$ *satisfies* $\gamma R \ge 1$*, then*

$$\big| \langle X\Omega, A\Omega\rangle - \langle X\Omega, \Omega\rangle\langle \Omega, A\Omega\rangle \big| \le 2\, \|X\Omega\|\, \|A\|\, e^{-\gamma R/2}. \tag{23}$$

Proof. Put $P_0 = |\Omega\rangle\langle\Omega|$ and define

$$F(t) := \langle X\Omega, (e^{itH} - P_0)A\Omega\rangle, \qquad G(t) := \langle A^*\Omega, (e^{-itH} - P_0)X\Omega\rangle.$$

Step 1: spectral supports and variations. Inserting the spectral resolution of H on $(1-P_0)\mathcal{H}$ and using (G_γ), F and G are Fourier transforms of complex measures supported in $[\gamma,\infty)$ and $(-\infty,-\gamma]$ respectively, with total variations at most $\mathfrak{m} := \|X\Omega\|\,\|A\|$ (Cauchy–Schwarz on the polar decomposition of the spectral measures).

Step 2: locality identity. For $|t| < R$: by Proposition 2.7, $\beta_t(A) \in \mathfrak{A}(O_{|t|})$, and $\operatorname{dist}(B, O_{|t|}) \geq R - |t| > 0$, so there is a bounded open $U \supseteq B$ disjoint from $O_{|t|}$. Affiliation of X with $\mathfrak{A}(U)$ implies that $\beta_t(A) \in \mathfrak{A}(O_{|t|}) \subseteq \mathfrak{A}(U)'$ preserves $D(X)$ and commutes with X on it. Since $e^{-itH}\Omega = \Omega$,

$$\langle X\Omega, e^{itH}A\Omega\rangle = \langle \Omega, X\beta_t(A)\Omega\rangle = \langle \Omega, \beta_t(A)X\Omega\rangle = \langle A^*\Omega, e^{-itH}X\Omega\rangle.$$

The P_0 terms agree because $\langle \Omega, X\Omega\rangle$ is real ($X = X^*$, $\Omega \in D(X)$). Hence $F = G$ on $(-R, R)$.

Step 3: conclusion. Lemma 9.2 gives $|F(0)| \leq 2\mathfrak{m}\, e^{-\gamma R/2}$, and $F(0) = \langle X\Omega, A\Omega\rangle - \langle X\Omega, \Omega\rangle\langle\Omega, A\Omega\rangle$. □

10 The one-path LPPL estimate

The setting of §8 remains in force: $g, g' \in \mathcal{G}$ equal to 1 on a common nonempty open interval, $g_s = (1-s)g + sg'$, $v = g' - g$; note $|v| \leq 1$.

Lemma 10.1 (state derivative)**.** *For self-adjoint $A \in B(\mathcal{H})$ and $0 < s < 1$,*

$$\frac{d}{ds}\,\omega_{g_s}(A) = -2\,\mathrm{Re}\int_{\mathbb{R}} W_\gamma(t)\Big[\langle V(v)\Omega_s, \beta_t^{g_s}(A)\Omega_s\rangle - \langle V(v)\Omega_s, \Omega_s\rangle\,\omega_{g_s}(A)\Big]\,dt, \tag{24}$$

and the integral converges absolutely.

Proof. Since A is bounded and self-adjoint, Proposition 8.2(ii) gives

$$\frac{d}{ds}\,\omega_{g_s}(A) = 2\,\mathrm{Re}\langle \dot\Omega_s, A\Omega_s\rangle = -2\,\mathrm{Re}\langle V(v)\Omega_s, H_s^{-1}Q_sA\Omega_s\rangle,$$

using (15) and self-adjointness of $H_s^{-1}Q_s$. Insert the exact filter (18) for $H = H_s$ (admissible by (G_γ)), applied to the vector $Q_sA\Omega_s$: since $e^{itH_s}\Omega_s = \Omega_s$,

$$e^{itH_s}Q_sA\Omega_s = e^{itH_s}\big(A\Omega_s - \omega_{g_s}(A)\Omega_s\big) = \beta_t^{g_s}(A)\Omega_s - \omega_{g_s}(A)\Omega_s,$$

which yields (24). The integral is absolutely convergent because $W_\gamma \in L^1$ (Lemma 9.1) and the bracket is bounded by $2\|A\|\,\|V(v)\Omega_s\|$; here $V(v)\Omega_s$ exists by Proposition 2.6 (v has compact support). □

Lemma 10.2 (localization of the perturbation)**.** *Choose a smooth partition of unity $\sum_{j\in\mathbb{Z}}\eta_j = 1$ with $0 \leq \eta_j \leq 1$ and $\operatorname{supp}\eta_j \subset (j-1, j+1)$, and set $h_j := \eta_j v$, $V_j := V(h_j)$, $B_j := \operatorname{supp} h_j$. Then only finitely many h_j are nonzero,*

$$V(v)\xi = \sum_j V_j\xi \quad \text{on common domains}, \qquad \|V_j\Omega_s\| \leq M \quad (0 \leq s \leq 1). \tag{25}$$

Proof. Finitely many h_j are nonzero because v has compact support; the finite additivity is the convention of §2.1. Each h_j is real and measurable with $\|h_j\|_\infty \le \|\eta_j\|_\infty \|v\|_\infty \le 1$ and $\operatorname{supp} h_j \subseteq [j-1, j+1]$, an interval of length 2. Since $g_s \in \mathcal{C}_{\mathrm{H}}$ (Lemma 3.1) and $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$, Theorem 5.1 applies to the cutoff g_s and gives $\|V_j \Omega_s\| \le M$. □

Lemma 10.3 (two-regime bound)**.** *Let $A = A^* \in \mathfrak{A}(O)$ with $\|A\| \le 1$, and for a nonzero h_j put $r_j := \operatorname{dist}(B_j, O)$ and*

$$C_{j,s}(t) := \langle V_j \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle - \langle V_j \Omega_s, \Omega_s \rangle \, \omega_{g_s}(A).$$

If $\gamma r_j/2 \ge 1$, then

$$|C_{j,s}(t)| \le \begin{cases} 2M e^{-\gamma r_j/4}, & |t| \le r_j/2, \\ 2M, & t \in \mathbb{R}. \end{cases} \tag{26}$$

Proof. For arbitrary t, bound each of the two terms by $\|V_j \Omega_s\| \, \|A\| \le M$ (Lemma 10.2), giving $2M$. For $|t| \le r_j/2$: Proposition 2.7 places $\beta_t^{g_s}(A)$ in $\mathfrak{A}(O_{|t|})$, and $\operatorname{dist}(B_j, O_{|t|}) \ge r_j - |t| \ge r_j/2$, with $\gamma \cdot \operatorname{dist}(B_j, O_{|t|}) \ge \gamma r_j/2 \ge 1$. Apply Lemma 9.3 with $H = H(g_s)$, $X = V_j$ (affiliated with every $\mathfrak{A}(U)$, $U \supseteq B_j$, by §2.1; $\Omega_s \in D(V_j)$ by Proposition 2.6), and the observable $\beta_t^{g_s}(A) \in \mathfrak{A}(O_{|t|})$; note $\langle \Omega_s, \beta_t^{g_s}(A) \Omega_s \rangle = \omega_{g_s}(A)$ by invariance of Ω_s. This gives

$$|C_{j,s}(t)| \le 2 \|V_j \Omega_s\| \, \|\beta_t^{g_s}(A)\| \, e^{-\gamma \operatorname{dist}(B_j, O_{|t|})/2} \le 2M e^{-\gamma r_j/4}. \qquad \square$$

Proposition 10.4 (one-path estimate)**.** *Let $d \ge d_0 := 4 + 2/\gamma$ and suppose $g \equiv g' \equiv 1$ on O_d. Then*

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le 32 M \sum_{n \ge \lfloor d \rfloor} \Big(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \Big). \tag{27}$$

Proof. Step 1: all perturbation cells are far from O. On O_d, $v = g' - g = 0$, and O_d is open, so $\operatorname{supp} v \subseteq \mathbb{R} \setminus O_d$; hence every nonzero h_j has $r_j = \operatorname{dist}(B_j, O) \ge d \ge d_0$. In particular $\gamma r_j/2 \ge \gamma d_0/2 = 2\gamma + 1 \ge 1$, so Lemma 10.3 applies to every j.

Step 2: derivative bound. Fix $A = A^* \in \mathfrak{A}(O)$, $\|A\| \le 1$. By Lemmas 10.1 and 10.2,

$$\frac{d}{ds} \omega_{g_s}(A) = -2 \operatorname{Re} \sum_j \int_{\mathbb{R}} W_\gamma(t) \, C_{j,s}(t) \, dt.$$

Split each t-integral at $|t| = r_j/2$ and use (26): the inner region contributes at most $\|W_\gamma\|_1 \cdot 2M e^{-\gamma r_j/4}$, the outer at most $T_\gamma(r_j/2) \cdot 2M$. With the overall factor 2,

$$\Big| \frac{d}{ds} \omega_{g_s}(A) \Big| \le 4M \sum_j \Big(\|W_\gamma\|_1 e^{-\gamma r_j/4} + T_\gamma(r_j/2) \Big). \tag{28}$$

Step 3: shell count. At most eight indices j have $r_j \in [n, n+1)$, for any integer $n \ge \lfloor d \rfloor$. Indeed, write $O = (a, b)$. Since $B_j \subseteq [j-1, j+1]$ has length ≤ 2, avoids O_d, and O_d is an interval of length $> 2d \ge 8$, the set B_j lies wholly on one side of O_d. On the right side, compactness of B_j gives $x_j \in B_j$ with $x_j - b = r_j$; then $|x_j - j| \le 1$ and $r_j \in [n, n+1)$ force $j \in [b + n - 1, b + n + 2]$, an interval containing at most four integers. The left side contributes at most four more.

Step 4: integrate and sum shells. Both functions of r_j in (28) are decreasing in r_j, so grouping the indices into the shells of Step 3 gives

$$\Big| \frac{d}{ds} \omega_{g_s}(A) \Big| \le 4M \cdot 8 \sum_{n \ge \lfloor d \rfloor} \Big(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \Big).$$

Integrate over $s \in [\varepsilon, 1-\varepsilon]$, where differentiability holds (Proposition 8.2), and let $\varepsilon \downarrow 0$ using the endpoint norm continuity of $s \mapsto P_s$; this bounds $|\omega_g(A) - \omega_{g'}(A)|$ by the right side of (27). Finally, $\omega_g - \omega_{g'}$ is a hermitian functional, so its norm on $\mathfrak{A}(O)$ is the supremum of $|\omega_g(A) - \omega_{g'}(A)|$ over self-adjoint contractions $A \in \mathfrak{A}(O)$. □

Lemma 10.5 (the rate function is super-polynomial)**.** *Define*

$$\Psi_\gamma(d) := \begin{cases} 32M \displaystyle\sum_{n \geq \lfloor d \rfloor} \Big(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2) \Big), & d \geq d_0, \\ \max\Big\{ 2,\ \textit{the same expression} \Big\}, & 0 \leq d < d_0. \end{cases} \tag{29}$$

Then for every $N \geq 1$ there is $C_{N,\gamma,M} < \infty$ with

$$\Psi_\gamma(d) \leq C_{N,\gamma,M}\, (1+d)^{-N} \qquad (d \geq 0). \tag{30}$$

In particular $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_\gamma(d)$ whenever $g \equiv g' \equiv 1$ on O_d, $d \geq 0$.

Proof. For $d \geq d_0$: the exponential part sums to $\|W_\gamma\|_1 e^{-\gamma\lfloor d \rfloor/4}(1 - e^{-\gamma/4})^{-1} \leq C'_\gamma e^{-\gamma d/4}$, which is $O_N((1+d)^{-N})$ for every N. For the second part choose $q = N+2$ in the tail bound of Lemma 9.1(i): $T_\gamma(n/2) \leq C_{q,\gamma}(1+n/2)^{-q} \leq C_{q,\gamma} 2^q (1+n)^{-q}$, and

$$\sum_{n \geq \lfloor d \rfloor} (1+n)^{-N-2} \leq \int_{\lfloor d \rfloor - 1}^{\infty} (1+x)^{-N-2}\, dx = \frac{\lfloor d \rfloor^{-N-1}}{N+1} \leq \frac{C''_N}{(1+d)^{N+1}},$$

using $\lfloor d \rfloor \geq d - 1 \geq \frac{3}{5}(1+d)$ for $d \geq d_0 \geq 4$. For $0 \leq d < d_0$: Ψ_γ is nonincreasing there, so the left side of (30) is bounded by $\Psi_\gamma(0) = \max\{2,\, 32M \sum_{n \geq 0}(\|W_\gamma\|_1 e^{-\gamma n/4} + T_\gamma(n/2))\} < \infty$, while the right side is bounded below by $C_{N,\gamma,M}(1+d_0)^{-N}$; enlarging $C_{N,\gamma,M}$ covers this range. The final assertion combines Proposition 10.4 with the trivial bound $\|\omega_g - \omega_{g'}\| \leq 2$ for $d < d_0$. □

11 Convergence of the full net; local normality

From here on the abstract gap parameter is instantiated: by Theorem 7.8, (G_γ) holds with $\gamma = m_1$, and we write $\Psi_{m_1} := \Psi_\gamma|_{\gamma = m_1}$ with M the constant of Theorem 5.1.

Theorem 11.1 (full-net convergence with rate)**.** *Let $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. There is a state ω_∞ on $\mathfrak{A}$ such that, for every bounded open interval O,*

$$\lim_{g \in \mathcal{G}} \big\| (\omega_g - \omega_\infty)|_{\mathfrak{A}(O)} \big\| = 0,$$

and

$$\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d) \qquad \textit{whenever } g \equiv 1 \textit{ on } O_d. \tag{31}$$

Proof. Step 1: the net is locally Cauchy. Fix O and $\varepsilon > 0$. By Lemma 10.5, choose $d \geq d_0$ with $\Psi_{m_1}(d) < \varepsilon$, and choose $g_0 \in \mathcal{G}$ whose plateau interior contains O_d. If $g, g' \succcurlyeq g_0$, then $\{g = 1\}^\circ \supseteq \{g_0 = 1\}^\circ \supseteq O_d$ and likewise for g'; in particular $g \equiv g' \equiv 1$ on O_d and their common plateau interior is nonempty, so Proposition 10.4 (with $\gamma = m_1$) applies and gives

$$\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \leq \Psi_{m_1}(d) < \varepsilon. \tag{32}$$

Step 2: limit functionals and compatibility. Each dual $\mathfrak{A}(O)^*$ is complete, so (32) produces a norm limit φ_O of $(\omega_g|_{\mathfrak{A}(O)})_{g \in \mathcal{G}}$. If $O_1 \subseteq O_2$, then restriction to $\mathfrak{A}(O_1)$ of the $\mathfrak{A}(O_2)$-limit

is again the norm limit of $\omega_g|_{\mathfrak{A}(O_1)}$, hence equals φ_{O_1} by uniqueness of norm limits. The compatible family $(\varphi_O)_O$ defines a linear functional ω_∞ on the norm-dense union $\bigcup_O \mathfrak{A}(O)$, with $|\omega_\infty(A)| \le \|A\|$, $\omega_\infty(1) = 1$, and $\omega_\infty(A^*A) = \lim_g \omega_g(A^*A) \ge 0$ for local A; it extends uniquely by norm continuity to $\mathfrak{A}$, and the extension is a state (a norm-one positive unital functional on a norm-dense subalgebra extends to a state).

Step 3: the rate. Fix $g \equiv 1$ on O_d. The subset $\{g' \in \mathcal{G} : g' \equiv 1 \text{ on } O_d\}$ is cofinal in $\mathcal{G}$ (every plateau can be enlarged to contain O_d). For every such g', Proposition 10.4 gives $\|(\omega_g - \omega_{g'})|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ (via Lemma 10.5 for $d < d_0$); letting g' run through this cofinal subset, $\omega_{g'} \to \omega_\infty$ in the norm of $\mathfrak{A}(O)^*$, and the bound passes to the limit, proving (31). □

Proposition 11.2 (local normality)**.** *$\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every bounded open interval O.*

Proof. Let $0 \le A_i \uparrow A$ be a bounded increasing net in $\mathfrak{A}(O)$ with supremum A. Given $\delta > 0$, choose $g \in \mathcal{G}$ with $\|(\omega_g - \omega_\infty)|_{\mathfrak{A}(O)}\| < \delta$ (Theorem 11.1). Then

$$0 \le \omega_\infty(A) - \omega_\infty(A_i) = \big[\omega_\infty - \omega_g\big](A - A_i) + \omega_g(A - A_i) \le 2\delta\|A\| + \omega_g(A - A_i),$$

using $\|A - A_i\| \le 2\|A\|$. The cutoff state ω_g is a vector state, hence normal on the von Neumann algebra $\mathfrak{A}(O)$, so $\omega_g(A - A_i) \to 0$ along the net. Therefore $\limsup_i[\omega_\infty(A) - \omega_\infty(A_i)] \le 2\delta\|A\|$; let $\delta \downarrow 0$. Thus $\omega_\infty(A_i) \uparrow \omega_\infty(A)$, which is normality on $\mathfrak{A}(O)$. □

12 Ground state, invariances, and the Main Theorem

Proposition 12.1 (exact spectral passage)**.** *If $(g_n) \subseteq \mathcal{G}$ satisfies $d(g_n, O) \to \infty$ for every bounded interval O, and $\omega_{g_n} \to \omega$ in norm on every $\mathfrak{A}(O)$, then ω is a ground state of $(\mathfrak{A}, \alpha)$ in the spectral sense: $\omega \circ \alpha_t = \omega$ for all t; the function $t \mapsto \omega(A\alpha_t(B))$ is continuous and bounded for all $A, B \in \mathfrak{A}$; and*

$$\int_{\mathbb{R}} f(t)\, \omega(A\alpha_t(B))\, dt = 0$$

for every $f \in L^1(\mathbb{R})$ whose transform $\check{f}(E) = \int f(t)e^{itE}\, dt$ vanishes on $[0, \infty)$.

Proof. Exact identity. Fix a bounded interval O, $A, B \in \mathfrak{A}(O)$, $\mathfrak{m} := \|A\|\|B\|$, and $g \in \mathcal{G}$; let $\rho_g(\cdot) := \langle A^*\Omega_g, P_{H(g)}(\cdot)B\Omega_g\rangle$, a complex measure with $\operatorname{supp}\rho_g \subseteq \operatorname{spec} H(g) \subseteq [0, \infty)$ and $|\rho_g|(\mathbb{R}) \le \mathfrak{m}$, and $G_g(t) := \int e^{itE}\, d\rho_g(E)$. For $|t| < d(g, O)$, the cutoff equals one on $O_{|t|}$, so by (P) and $e^{-itH(g)}\Omega_g = \Omega_g$,

$$\omega_g\big(A\alpha_t(B)\big) = \langle \Omega_g, Ae^{itH(g)}Be^{-itH(g)}\Omega_g\rangle = \langle A^*\Omega_g, e^{itH(g)}B\Omega_g\rangle = G_g(t).$$

Moreover $\int_{\mathbb{R}} f\, G_g\, dt = \int_{[0,\infty)} \check{f}\, d\rho_g = 0$ for the stated class of f (Fubini, justified by $|\rho_g|(\mathbb{R}) \le \mathfrak{m}$ and $f \in L^1$; $\check{f} = 0$ on the support).

Invariance. For $A \in \mathfrak{A}(O)$ and fixed t, take n so large that $d(g_n, O) > |t|$. Then (P) and $e^{-itH(g_n)}\Omega_{g_n} = \Omega_{g_n}$ give directly

$$\omega_{g_n}(\alpha_t(A)) = \langle \Omega_{g_n}, e^{itH(g_n)}Ae^{-itH(g_n)}\Omega_{g_n}\rangle = \omega_{g_n}(A).$$

Both $\alpha_t(A)$ and A belong to $\mathfrak{A}(O_{|t|})$, so norm convergence on $\mathfrak{A}(O_{|t|})$ passes the identity to the limit: $\omega(\alpha_t(A)) = \omega(A)$, and by density on $\mathfrak{A}$.

Continuity and vanishing. Fix $A, B \in \mathfrak{A}(O)$ and $T > 0$. For n with $d(g_n, O) > T$ and $|t| \le T$: $A\alpha_t(B) \in \mathfrak{A}(O_T)$ with norm $\le \mathfrak{m}$, so

$$\sup_{|t|\le T} \big|\omega(A\alpha_t(B)) - \omega_{g_n}(A\alpha_t(B))\big| \le \mathfrak{m}\, \big\|(\omega - \omega_{g_n})|_{\mathfrak{A}(O_T)}\big\| \longrightarrow 0 :$$

$t \mapsto \omega(A\alpha_t(B))$ is a locally uniform limit of the continuous functions G_{g_n}, hence continuous, bounded by $\mathfrak{m}$. For f as stated and $\varepsilon(T) := \int_{|t|>T} |f|$:

$$\int_{\mathbb{R}} f\,\omega(A\alpha_\cdot(B)) = \underbrace{\int_{|t|\le T} f\,[\omega-\omega_{g_n}](A\alpha_t(B))}_{\le \|f\|_1\mathfrak{m}\|(\omega-\omega_{g_n})|_{\mathfrak{A}(O_T)}\|} + \underbrace{\int_{|t|\le T} f\,G_{g_n}}_{=-\int_{|t|>T} f\,G_{g_n}} + \int_{|t|>T} f\,\omega(A\alpha_t(B)),$$

where the middle identity uses $\int_{\mathbb{R}} f\,G_{g_n}\,dt = 0$. Hence the modulus is at most $\|f\|_1\mathfrak{m}\|(\omega-\omega_{g_n})|_{\mathfrak{A}(O_T)}\| + 2\mathfrak{m}\,\varepsilon(T)$; let $n\to\infty$, then $T\to\infty$. Joint norm continuity in (A,B) extends the conclusion from local A, B to $\mathfrak{A}$. □

The Main Theorem now follows by assembly.

Theorem 12.2 (Main Theorem)**.** *Let $0 \le \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. Then there is a state ω_∞ on $\mathfrak{A}$ such that:*

(1) (full-net convergence with rate) *for every bounded open interval O, $\lim_{g\in\mathcal{G}} \|(\omega_g-\omega_\infty)|_{\mathfrak{A}(O)}\| = 0$, and $\|(\omega_g-\omega_\infty)|_{\mathfrak{A}(O)}\| \le \Psi_{m_1}(d)$ whenever $g\equiv 1$ on O_d, with Ψ_{m_1} decaying faster than every inverse power;*

(2) (local normality) *$\omega_\infty|_{\mathfrak{A}(O)}$ is normal for every O;*

(3) (ground state) *ω_∞ is a ground state of $(\mathfrak{A},\alpha)$ in the spectral sense;*

(4) (invariances) *$\omega_\infty\circ\tau_a = \omega_\infty$ for all $a\in\mathbb{R}$; and for even $\mathscr{P}$, $\omega_\infty\circ\theta = \omega_\infty$;*

(5) (spectra and local moment)

$$\operatorname{spec} H(g) \subseteq \{0\}\cup[m_1,\infty) \quad (g\in\mathcal{C}_{\mathrm{H}}), \qquad \dim\ker H(g) = 1,$$

and $\sup_{g\in\mathcal{C}_{\mathrm{H}}} \|V(h)\Omega_g\| \le M$ for real h with $\|h\|_\infty \le 1$ supported in an interval of length 2.

The weak-coupling estimates imported are two printed theorems of [7]: Theorem 1.1.7 (Source Fact 2.2) and Theorem 1.1.11 at $B = I$, $n = 0$, the cutoff-uniform moment estimate (U_{GJS}) of Assumption A8(vi) (imported as printed since 31 October 2026; Scope paragraph there). Appendix C supplies a source-mechanism audit and analytic prerequisites but is not an independent proof of the cluster expansion. In addition, the proof uses the separately identified standard cutoff-construction inputs of Assumptions A1–A7. Proposition 3.2 verifies that the dimensionless weak-coupling hypothesis implies the source smallness condition. The theorem is an implication from exactly this list of printed statements; no other external input is used.

Proof. (5) is Theorem 7.8 and Theorem 5.1. These supply (G_γ) with $\gamma = m_1$ and (M_2), so Theorem 11.1 and Proposition 11.2 give (1)–(2). For (3): choose $\widetilde{g}_n \in \mathcal{G}$ with $\widetilde{g}_n \equiv 1$ on $(-n,n)$; then $d(\widetilde{g}_n, O)\to\infty$ for every O and, by (1), $\omega_{\widetilde{g}_n}\to\omega_\infty$ in norm on every $\mathfrak{A}(O)$; Proposition 12.1 applies and gives (3) together with the α-invariance. For (4): translation is an order isomorphism of $\mathcal{G}$ with cofinal range, so by (T) and (1),

$$\omega_\infty(A) = \lim_g \omega_{\tau_a g}(A) = \lim_g \omega_g(\tau_{-a}A) = \omega_\infty(\tau_{-a}A)$$

for local A, and by density on $\mathfrak{A}$. The $\mathbb{Z}_2$ statement is the local norm limit of Lemma 2.8: $\omega_g(\theta(A)) = \omega_g(A)$ for $A\in\mathfrak{A}(O)$, and $\theta(A)\in\mathfrak{A}(O)$ (θ maps $W(f,h)$ to $W(-f,-h)$ and hence preserves each local algebra), so (1) passes the identity to ω_∞, and density extends it to $\mathfrak{A}$. □

Corollary 12.3 (the Glimm–Jaffe vacuum is canonical at weak coupling)**.** *Every w^* limit point of the (averaged or unaveraged) finite-volume ground states constructed in [3, §2] coincides on each $\mathfrak{A}(O)$ with ω_∞; in particular the construction of [3] is subsequence- and averaging-independent in the weak-coupling region.*

Proof. For the unaveraged net this is Theorem 12.2(1). For the averaged states of [3, (2.3)–(2.5)]: fix $g \in \mathcal{G}$ with $g \equiv 1$ on $[-3,3]$ and set $g_n := g(\cdot/n) \in \mathcal{G}$; let $\sigma_{a'}$ denote the space-translation automorphisms used there (our $\tau_{\epsilon a'}$, with $\epsilon \in \{\pm 1\}$ the direction convention), and let $\bar{h} \geq 0$, $\operatorname{supp}\bar{h} \subseteq [-1,1]$, $\int \bar{h} = 1$, be the averaging weight. Then

$$\begin{aligned}\omega_n(A) &= \frac{1}{n}\int \langle \Omega_{g_n}, \sigma_{a'}(A)\Omega_{g_n}\rangle\, \bar{h}(a'/n)\, da' = \int_{-1}^{1} \bar{h}(u)\, \omega_{g_n}\big(\tau_{\epsilon n u}A\big)\, du \\ &= \int_{-1}^{1} \bar{h}(u)\, \omega_{\tau_{-\epsilon n u} g_n}(A)\, du,\end{aligned}$$

by (T) (the estimate below is invariant under $\epsilon \to -\epsilon$). Since $g_n \equiv 1$ on $[-3n, 3n]$, the shifted cutoff $\tau_{-\epsilon n u} g_n \in \mathcal{G}$ equals 1 on an interval containing $[-2n, 2n]$ for every $|u| \leq 1$; hence for $O \subseteq [-R, R]$, $d(\tau_{-\epsilon n u} g_n, O) \geq 2n - R$ *uniformly in* u, and Theorem 12.2(1) gives

$$\|(\omega_n - \omega_\infty)|_{\mathfrak{A}(O)}\| \leq \int_{-1}^{1} \bar{h}(u)\, \big\|(\omega_{\tau_{-\epsilon n u} g_n} - \omega_\infty)|_{\mathfrak{A}(O)}\big\|\, du \leq \Psi_{m_1}(2n - R) \longrightarrow 0.$$

No translation invariance of ω_∞ is needed for this bound. □

13 The two-phase dichotomy

The uniform gap of Theorem 7.8 is incompatible with a fully classified two-phase structure. The incompatibility is stated as a conditional theorem: *all four* hypotheses below are load-bearing, and (B4) in particular cannot be omitted — dropping it was the one proof-breaking defect (P1) found by the independent audit of the predecessor manuscript (Appendix B).

Hypothesis 13.1 (structure of a broken phase)**.** *$\mathscr{P}$ is even, and:*

(B1) *there are translation-invariant ground states ω_+ and ω_- of $(\mathfrak{A}, \alpha)$ such that, for all bounded local X, Y,*

$$\omega_\pm(X\tau_a(Y)) - \omega_\pm(X)\,\omega_\pm(\tau_a(Y)) \longrightarrow 0 \qquad (|a| \to \infty);$$

(B2) *$\omega_+ \circ \theta = \omega_-$ and $\omega_- \circ \theta = \omega_+$;*

(B3) *there are a bounded local observable $B = B^* \in \mathfrak{A}(O)$ and $m \neq 0$ such that $\omega_\pm(B) = \pm m$;*

(B4) *every θ-invariant ground state of $(\mathfrak{A}, \alpha)$ equals $\tfrac{1}{2}(\omega_+ + \omega_-)$.*

Theorem 13.2 (conditional exclusion of a two-phase structure)**.** *Assume that $\mathscr{P}$ is even and $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$. Then the assertions* (B1)–(B4) *of Hypothesis 13.1 cannot hold simultaneously.*

Proof. Assume that $\mathscr{P}$ is even and (B1)–(B4) hold. Choose $\widetilde{g}_n \in \mathcal{G}$ with $\widetilde{g}_n \equiv 1$ on $(-n, n)$ and write $\omega_n := \omega_{\widetilde{g}_n}$. Fix $a \in \mathbb{R}$ such that

$$R_a := \operatorname{dist}(O, O + a) > 0, \qquad m_1 R_a \geq 1,$$

where O is the localization interval of the order parameter B from (B3).

Step 1: uniform cutoff clustering for the bounded order parameter. For the cutoff Hamiltonian $H(\widetilde{g}_n)$ apply Lemma 9.3 with $\gamma = m_1$ (Theorem 7.8), unbounded-argument slot $X = B$, bounded-argument slot $A = \tau_a(B) \in \mathfrak{A}(O + a)$, and localization set $\overline{O}$. The bounded operator B has domain $\mathcal{H}$; because $B \in \mathfrak{A}(O) \subseteq \mathfrak{A}(U)$ for every bounded open $U \supseteq \overline{O}$, it is affiliated with every such $\mathfrak{A}(U)$, and $\operatorname{dist}(\overline{O}, O + a) = R_a$. Thus all hypotheses of that lemma hold, and since $\|B\Omega_{\widetilde{g}_n}\| \le \|B\|$ and $\|\tau_a(B)\| = \|B\|$, it gives, uniformly in n,

$$\big|\omega_n\big(B\,\tau_a(B)\big) - \omega_n(B)\,\omega_n(\tau_a(B))\big| \le 2\|B\|^2 e^{-m_1 R_a/2}. \tag{33}$$

Step 2: passage to the limit. For fixed a, all three observables in (33) belong to the local algebra of one bounded interval containing $O \cup (O + a)$. The local-norm convergence of Theorem 12.2(1) therefore permits passage to $n \to \infty$ in (33); translation invariance of ω_∞ (Theorem 12.2(4)) then yields

$$\big|\omega_\infty\big(B\,\tau_a(B)\big) - \omega_\infty(B)^2\big| \le 2\|B\|^2 e^{-m_1 R_a/2} \xrightarrow[|a|\to\infty]{} 0. \tag{34}$$

Step 3: the midpoint identification. By Theorem 12.2(3)–(4), ω_∞ is a θ-invariant ground state of $(\mathfrak{A}, \alpha)$. Assumption (B4) consequently gives

$$\omega_\infty = \tfrac{1}{2}(\omega_+ + \omega_-), \qquad \omega_\infty(B) = \tfrac{1}{2}(m - m) = 0,$$

using (B3).

Step 4: long-range order of the midpoint. By (B1), translation invariance and clustering of each $\omega_\pm$ imply

$$\omega_\pm\big(B\,\tau_a(B)\big) - \omega_\pm(B)\,\omega_\pm(\tau_a(B)) \longrightarrow 0, \qquad \omega_\pm(\tau_a(B)) = \omega_\pm(B) = \pm m.$$

Hence

$$\omega_\infty\big(B\,\tau_a(B)\big) = \tfrac{1}{2}\big(\omega_+\big(B\,\tau_a(B)\big) + \omega_-\big(B\,\tau_a(B)\big)\big) \longrightarrow \tfrac{1}{2}(m^2 + m^2) = m^2 \neq 0,$$

contradicting (34), whose left side tends to zero because $\omega_\infty(B) = 0$. □

Remark 13.3 (scope of the phase-transition literature)**.** The GJS phase-transition and mean-field papers [8, 9] prove the existence of phase transitions and of two pure phases in specified strong-coupling regimes. As cited here, they are context: they do not, by themselves, supply the full Hamiltonian phase-space classification (B1)–(B4) — in particular not the midpoint identification (B4) — for the state space of $(\mathfrak{A}, \alpha)$. This manuscript therefore makes no unconditional assertion that its state space carries the structure of Hypothesis 13.1; Theorem 13.2 says only that, at weak coupling, that structure cannot coexist with the cutoff-uniform gap. The two rigorous options recorded by the audit — state (B1)–(B4) explicitly, or delete the corollary — are resolved here by the first.

14 Discussion

Relation to GRS Problems 1–2. Problem 2 of [10, p. 126] asks for local convergence of Euclidean Gibbs states; Theorem 12.2(1) is a Hamiltonian counterpart, in norm on local von Neumann algebras, for the vacuum family. Problem 1 asks for uniform local L^p bounds on the Euclidean vacuum density. The note added on [10, p. 128] records Newman's solution of the small-coupling $p = 1$ versions, so the printed problems must not be read as an unqualified current open-problem claim. The present proof neither establishes nor needs Problem 1: the

identification step uses the exact spectral identity of Proposition 12.1, and the convergence step uses the Cauchy estimate.

Sharpness. In the exactly solvable quadratic model the true rate is $\mathrm{poly}(d)\, e^{-2m_1 d}$ (companion note `m2/`: a round trip through the region where the two Hamiltonians differ, at the local Agmon mass). Ψ_{m_1} is super-polynomial but sub-exponential. The present filter argument does not obtain the sharp exponential rate; a proof of such a rate would require an additional resolvent or Combes–Thomas input not supplied here.

Uniqueness. ω_∞ is canonical for the cutoff family (Corollary 12.3), and translation invariant without averaging. The theorem does not identify every locally normal translation-invariant ground state of $(\mathfrak{A}, \alpha)$ with ω_∞. In particular, the Euclidean results [13, 14] do not by themselves provide the reverse reconstruction and state-space identification needed for that Hamiltonian conclusion; companion note `m3/` records that missing implication as its item (R3).

Beyond weak coupling. Theorem 13.2 is conditional on Hypothesis 13.1. Under exactly those hypotheses, a proof based on a cutoff-uniform gap cannot extend to that two-phase regime. The phase-transition results [8, 9], as cited here, are not used as a substitute for (B4) (Remark 13.3), and this manuscript makes no unconditional assertion that its Hamiltonian state space has the classification (B1)–(B4).

Part II. Uniformity in a lattice/wavelet resolution — the operator-algebraic renormalization programme over the maps of [18] — is a separate problem with no mathematical dependence in either direction, and its record lives in `part_ii/` and `strategy.md`. Its present state, in one paragraph: for the free nearest-neighbour model, with lattice vacua $\omega_N^{(0)}$ and continuum vacuum $\omega^{(0)}$, there are local Weyl unitaries with $\liminf_{N\to\infty} \|\omega_N^{(0)} - \omega^{(0)} \circ \beta_N\|_{\mathfrak{W}_N(O)} \geq e^{-\sqrt{2+\sqrt{2}}/4} - e^{-3\pi/16} > 0$ for the Daubechies observable embedding β_N, so no diagonal (finest-scale) local-state-norm error $C\varepsilon_N^\vartheta$, $\vartheta > 0$, is possible, and the meaningful comparison is at fixed coarse scale; there, the free full sequence converges on each fixed coarse Weyl generator with an explicit power rate, an interacting fixed-torus projective cluster state exists unconditionally along a cofinal subnet, and the full-sequence identified interacting theorem is reduced to a single scalar characteristic-functional limit, which remains open.

Standing scope

Everything in this manuscript is restricted to the weak-coupling region $0 \leq \lambda/m_0^2 < \varepsilon_{\mathscr{P}}$ of §2.6; no claim is made for the full noncritical region. The external residue — a second-reader verification of the GJS page images — is recorded in Appendix B.

A Constants

All values below are fixed by the printed norms (1) (gate R2, page-image reading) and the displays of the body sections; each row was re-verified against its defining display at gates G3a/G3b (see `part_i/README.md`).

constant	value / definition	defined in
K_1, K_2, C_7, C_8, m_1	GJS constants; normalization $K_2 \geq 1$	Assumption A8
$\varepsilon_{\mathscr{P}}$	λ_*/M_*^2	§2.6
N_I	$\leq \lfloor L \rfloor + 2 \leq L + 2$ ($\lvert I \rvert = L$)	Lemma 6.1
$\mathfrak{K}(p')$	$K_1 K_2^{8p'} (4p')!$	Thm. 5.1
M^2	$9\, C_8\, \mathfrak{K}(p')\, \lambda^2 \big(\sum_{n=0}^{2p'} \lvert a_n \rvert\big)^2$	Thm. 5.1
M^2 (φ^4)	$9 \cdot 8!\, C_8 K_1 K_2^{16} \lambda^2 = 362\,880\, C_8 K_1 K_2^{16} \lambda^2$	Cor. 5.2
K'	$K_2^2 N_I^{1/2} \lVert f \rVert_2$ (exponential-moment base)	Lemma 7.6
$\widehat{C}_Q$	$\max\{C_7 e^{2m_1} \lVert Q \rVert_{\mathrm{GJS}}^2,\ e^{3m_1} \lVert \psi_Q \rVert^2\}$	Prop. 7.5
$\mathfrak{m}$	total-variation bound, separation lemma	Lemma 9.2
d_0	$4 + 2/\gamma$	Prop. 10.4
shell count	≤ 8 indices j with $r_j \in [n, n+1)$	Prop. 10.4
$\Psi_\gamma(d)$	$32M \sum_{n \geq \lfloor d \rfloor} \big(\lVert W_\gamma \rVert_1 e^{-\gamma n/4} + T_\gamma(n/2) \big)$; $\max\{2, \cdot\}$ if $d < d_0$	Lemma 10.5
Ψ_γ tail	$\leq C_{N,\gamma,M} (1+d)^{-N}$ for every N	Lemma 10.5

B Provenance, corrections history, and dossier

Frozen sources of this manuscript

file	role
`m6/main.tex` (v2, audited repair)	predecessor manuscript; statements and compressed proofs
`audit_2026/critical_proof_audit.tex`	independent audit (frozen against the v1 SHA-256)
`audit_2026/lamport_euclidean_to_gap.tex`	fully explicit proof, GJS $\Rightarrow$ (M_2)+(G)
`audit_2026/lamport_gap_to_main.tex`	fully explicit proof, (M_2)+(G) $\Rightarrow$ main theorem
`audit_2026/citation_screenshot_dossier.tex` + `evidence/`	page images for every import
`m4/r1-check.tex`	gate R1: h-uniformity of the Euclidean inputs
`m4/r2-check.tex`	gate R2: printed norms and exact constants (page image, p. 594)
`m0/m0.tex`, `m2/m2.tex`, `m3/m3.tex`	spectral passage; quadratic-model rate; dichotomy
`m4/m2-audit.tex`, `m4/g-side.tex`	superseded derivations (with update notices)
`strategy.md`	programme record, incl. the unification plan and gates

The two structured proof reconstructions follow the hierarchical proof format of [19].

Corrections history (chronological)

- *v0 → v1 merge* (9 Aug 2026): filter-transform sign corrected to $+1/E$; spurious factor $\frac{1}{2}$ removed from the one-path bound; an unnecessary g'' detour eliminated; the averaging identity of the canonical-vacuum corollary repaired.
- *Gate R2* (9 Aug 2026, page-image inspection of [7, p. 594]): prefactor K_1 to the first power; local weight $K_2^{2N(\Delta)}$; the earlier "correction" $K_2^{4p} \to K_2^{2p}$ of `m4/m2-audit.tex`, Rem. 5.3(i), was an OCR artifact and is withdrawn; a half-degree slip in the v0 assembly ($(2p')!$, $K_2^{4p'}$ in place of $(4p')!$, $K_2^{8p'}$) found and fixed. The cell-pair count for (M_2) is 9, not 32.

- *Independent audit* (9 Aug 2026): severity register P1 (two-phase corollary missing premise (B4) — repaired by conditioning on Hypothesis 13.1); S1–S3 (citation pinpoints: FKN, hypercontractivity, domain smoothing — repaired in Assumptions A2, A4, A5); L1–L6 (cell count $\lfloor L \rfloor + 2$; negative-time norm ratio c_g^{-1}; nonnegative FKN inputs $F_1 = F_2 = |V(h)|$; signed/clipped truncations; exponential-moment prefactor $C_8 K_1$; spectral-parameter range $0 \le \beta < m_1$); C1 (dimensionless scaling reduction made explicit); C2 (all imports enumerated — Assumptions A1–A8); H1–H2 (literature claims narrowed; GRS $p = 1$ note recorded). No P0 defect was found.

- *Re-sourcing of A8(vi)* (31 October 2026, after the council record `ugjs/PLAN.md`): item (vi) is read off the printed Theorem 1.1.11 of [7] (p. 596, "uniformly in h") in the specialization $B = I$ ($T = 0$ on p. 600), $n = 0$ (named on p. 629), that the source prints as its proof of Theorem 1.1.8; the hypothesis clause naming U_{GJS} is removed from the statements of Theorem 12.2 and Proposition C.7; no proof changed, no label moved. In the same pass: item (ii) names the time-integrated monomials (1.1.6) and the square convention of p. 593; the Scope paragraph records the one case of Theorem 1.1.7 that its printed proof does not reach (sharp-time monomials across strips transverse to the time-zero line, unused here); the mechanism audit of Appendix C gains item M6 (the stability input (2.4.34) of GJS p. 624, which its items M1 and M3–M5 had wrongly declared absent) and its covariance description is corrected from "Dirichlet" to "decoupled".

- *Withdrawn routes* (kept in the record, not used here): `m4/h6-attack.tex` Thm. 2.3 (withdrawn; UV-divergent double commutator); gap-based (H6) proofs; wavelet-MRA-as-Polchinski decomposition (see Part II documents).

External residue

Audit closure item C5 — verification by a second human reader of the GJS page images — remains open at the level of this research record. The itemized reading checklist is `part_i/C5_reading_record.md`: items 1–4 (the cutoff class, the localized monomials and squares, the norms, Theorem 1.1.7 and the dilation; pp. 589, 590, 593, 594), item 9 (pp. 596–597, including (1.1.16) "uniformly in h"), and items 12–13 (added 31 October 2026: Theorem 1.1.11's "monomials of type (1.1.5)", "$B_T \dots$ at time $T > 0$" and the rotation sentence; p. 600's specialization and p. 628's hypotheses of Theorem 4.1) carry the imports of Assumption A8; items 10–11 (the two forms (1.1.5)–(1.1.6) with p. 595's multi-time notation; Theorem 1.1.7's strip without orientation) carry the disclosure of the Scope paragraph, point (b); items 5–8 are corroboration. Every status sentence of this manuscript is to be read with these human readings named as open beside it.

C h-uniformity: source audit and analytic prerequisites

This appendix has two sharply separated purposes. First, it records the audit of the source mechanism that preceded the re-sourcing of Assumption A8(vi) on 31 October 2026 (items M1–M5, with the correction M6 added that day). Second, it proves two h-sensitive analytic prerequisites over the full class $\mathcal{C}_{\mathrm{E}}$: normalization of dq_h and uniform L^p stability under the free measure. It does *not* reproduce the polymer expansion, its convergence summations, or the proof of all cluster-expansion constants. It therefore does not re-derive (U_{GJS}); that statement is imported as printed (Theorem 1.1.11 of [7] at $B = I$, $n = 0$; Assumption A8(vi)). Theorem 1.1.7's own statement makes its constants independent of h; its printed proof consumes Theorem 1.1.8 at the same h and reaches sharp-time monomials

only across strips parallel to the time-zero line (Scope paragraph of Assumption A8, point (b)).

Mechanism audit

The expansion machinery of [7, §§2–4] manipulates integrals $\int R(\Phi)\, B\, e^{-\mathcal{U}_h}\, d\mu_{0,C}$, where C is an interpolated covariance and $d\mu_{0,C}$ denotes the Gaussian measure with covariance C, through six devices; the following lists every point of contact with h (item M6 was added on 31 October 2026, after the council of `ugjs/PLAN.md` found it missing).

M1 (the norms are kernel norms). The norm (2.4.3) of an expansion term m_α is $\int du\, ds\, \exp[\cdots]\, \|w(\alpha, u, s, y, \cdot)\|_{L^2}$ times combinatorial weights (2.4.4)–(2.4.5) built from m_0, ε, K_2, $N(\Delta)$ (the source's script $\mathcal{K}_2$ of (2.4.4) is $e^{-(1-\varepsilon)m_0 \sum \delta n}$, not the constant K_2), times $K_1^\nu e^{\gamma t}$ ((2.4.3), p. 613): a functional of the *kernel* w alone. The norm itself contains no Gaussian expectation; the kernels w it is applied to are, however, defined through integrals of $e^{-V(Y_j)}$ under the interpolated measures ((2.3.4), (2.3.8), pp. 611–612), see M6. The norm-decrease Theorems 2.4.1, 3.1, 4.1 are inequalities between such kernel functionals, and their proofs (Lemmas 2.4.2–2.4.4, (3.2)–(3.6), §4) estimate kernels: contraction kernels δC against vertex kernels.

M2 (covariances are h-free). The interpolation $C(s) = sC' + (1-s)C''$ (2.0.1) is between covariances of the family generated from C_0 by "diagonalization" with respect to unions Y of unit squares, $C_Y = YCY + (1-Y)C(1-Y)$, and convex sums (p. 600); the kernels (2.2.2)–(2.2.3) are built from $(-d^2/dx^2 + m_0^2)$ and contour geometry. Verified against the legible scan (`m4/r1-check.tex`): no h enters any covariance, any contraction kernel, or any of the Propositions 2.1.x.

M3 (h in vertices, majorised at first contact). Vertices arise from functional derivatives of the spectator:

$$\frac{\delta}{\delta\Phi(x)} e^{-\mathcal{U}_h} = -\lambda\, h(x) :\mathscr{P}'(\Phi(x)): e^{-\mathcal{U}_h},$$

and iterates with $\mathscr{P}'', \ldots$ A vertex therefore contributes to the *kernel* a factor $\lambda h(x) p_j$ supported in a lattice square. The very first estimate applied to it is the printed one (p. 626): "$v(x)$ is $O(\lambda)$ times the characteristic function of a lattice square" — that is, $|\lambda h(x) p_j| \le O(\lambda)\chi_\Delta(x)$, using $0 \le h \le 1$ and nothing else. Downstream of this line the vertex kernels are h-free. This is one of two points where h touches a quantitative estimate; the other is M6.

M4 (the identities are exact and h-blind). The devices (2.0.4)–(2.0.5) (covariance interpolation), (2.2.1), (2.2.5), (2.2.7) (advancing annihilators), and (4.1)–(4.2), (4.5) (integration by parts) are identities applied to integrands $F = R(\Phi) B_T e^{-\mathcal{U}_h}$; $e^{-\mathcal{U}_h}$ rides along inside F. Their validity requires only that the integrals exist — integrability, not bounds. For the free-measure endpoint this is supplied, uniformly over $\mathcal{C}_{\mathrm{E}}$, by Theorem C.5; existence under the interpolated measures $d\mu_{0,C}$ is internal to the printed proof. A quantitative stability bound under those measures does enter the constants: see M6.

M5 (the final division adds no stability constant). The full expansion (4.3) reads

$$\int A B_T e^{-\mathcal{U}} d\mu_{0,C_0} = \sum_i \int \Big[\int dy\, w_i(y) :\Phi(y_1)\cdots\Phi(y_i):\Big] B_T e^{-\mathcal{U}} d\mu_{0,C_0} + \sum_r w_r \int e^{-\mathcal{U}} d\mu_{0,C_0},$$

with w_i, w_r *numerical kernels* controlled by the norm inequalities; the remainder terms are literally multiples of $\int e^{-\mathcal{U}} d\mu_0$ (their Table 1: final-r terms have "no vertices, no annihilators"). The proof of Theorems 1.1.8/1.1.11 then *divides (4.3) by* $\int e^{-\mathcal{U}} d\mu_{0,C_0}$ (p. 628): the first sum reassembles into dq_h-integrals of the comparison polynomials, and the remainder becomes $\sum_r w_r$ exactly. Every factor $e^{-\mathcal{U}_h}$ cancels against the normalisation. Consequently the final division introduces no stability constant; the stability bound that does enter the constants is the one of M6, upstream of the division.

M6 (stability under the interpolated measures; added 31 October 2026). The kernels w'' of (2.3.8) are bounded on p. 624 through a stability estimate under the interpolated measures: "For $m_0 \geq 1$, $\lambda \leq 1$, there is a constant K_3, independent of C, m_0 and λ such that

$$(2.4.34) \qquad 1 \leq \|e^{-V(Y)}\|_{L_2(d\Phi_C)} \leq e^{K_3|Y|}$$

and (2.4.35) [a bound on $\|P_l''(u,s,y,x_B)\|_{L_2(d\Phi_C)}$ by $\prod_\Delta K_3^{N(\Delta)} N(\Delta)!$ times an L_2 kernel norm]. These are standard estimates for such integrals (see [1])." The bound "yields (2.4.28), for $K_2 \geq K_3^2$" (p. 624), and Lemma 2.4.6 (p. 622) takes "$K_2 = K_2(m_0, K_3)$"; so K_3 enters every constant chosen after K_2 in the printed order "m_0, K_2, K_1, and then λ" (p. 616), C_7 and C_8 included. The cutoff sits inside $V(Y)$: (2.0.2) on p. 601 factors $V(h)$ as $V(Yh) + V((1-Y)h)$. Reference [1] is Dimock–Glimm, Adv. Math. 12 (1974), not consulted here; K_3 is printed independent of C, m_0 and λ, and h is not mentioned. Theorem C.5 does not supply (2.4.34): it is a bound under $d\mu_0$ for a fixed compact region, without linear growth in $|Y|$. The convergence argument on pp. 624–625 also uses $|m_\alpha| \leq \|m_\alpha\|_{2K_2} e^{O(\mathrm{Vol})}$, "where the volume of interaction, denoted Vol, is the area of the set of points within unit distance of suppt h. Here we use (2.4.34), (2.4.35) and a Schwarz inequality to factor out vertices" (p. 625), a per-h qualitative step. Items M1 and M3–M5, as transcribed from `m4/r1-check.tex` (Gate R1, 9 August 2026), declared this input absent; that statement was wrong about the source.

Proposition C.1 (consequence of the audited source mechanisms)**.** *Within mechanisms M1–M5 as recorded above, and excluding the stability input of M6, which lies outside them, every displayed quantitative bound is unchanged when h ranges over $0 \leq h \leq 1$: M3 replaces its only vertex occurrence by $|h| \leq 1$, while M4–M5 use identities or exact cancellation. Consequently the constants visible in those mechanisms depend on $(m_0, K_2, K_1, \lambda, n, \varepsilon, \mathscr{P})$ and not on h; the h-uniformity of K_3 in M6 is not among the audited mechanisms.*

Proof. By direct substitution in M1–M5: the quantitative chain displayed there is entirely at the level of kernel norms (M1), whose ingredients are h-free covariance/contraction kernels (M2) and vertex kernels majorised by $O(\lambda)\chi_\Delta$ (M3); the measure-level steps are exact identities (M4) or exact cancellations (M5). This proposition is deliberately limited to the audited mechanisms; it makes no completeness claim about portions of the source expansion not reproduced here. □

The two h-sensitive analytic facts, proved

Notation for this subsection: $G := (-\Delta_{\mathbb{R}^2} + m_0^2)^{-1}$ is the Euclidean covariance, $G(\zeta) = \frac{1}{2\pi} K_0(m_0|\zeta|)$; $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$, Λ compact; $\mathcal{U}_h = \lambda \int h : \mathscr{P}(\Phi) : dz$ and $Z(h) = \int e^{-\mathcal{U}_h} d\mu_0$ as in Assumption A3. Choose once and for all

$$\delta \in C_c^\infty(\mathbb{R}^2), \qquad \delta \geq 0, \qquad \delta(z) = \delta(-z), \qquad \operatorname{supp}\delta \subseteq \{z : |z| \leq 1\}, \qquad \int_{\mathbb{R}^2} \delta(z)\, dz = 1,$$

and, for $\kappa \geq 1$, define the two-dimensional rescaling $\delta_\kappa(z) := \kappa^2 \delta(\kappa z)$. Thus convolution by δ_κ is a positive average over a ball of radius κ^{-1}; a double convolution averages over displacements of radius at most $2\kappa^{-1}$. Set $\Phi_\kappa = \delta_\kappa * \Phi$, $c_\kappa(z) = \mathbb{E}[\Phi_\kappa(z)^2] \leq \frac{1}{2\pi} \log \kappa + C_*$ (translation invariance), and $\mathcal{U}_{h,\kappa} = \lambda \int h :\mathscr{P}(\Phi_\kappa):_{c_\kappa} dz$, Wick ordered with respect to the variance of the cutoff field, so that $\mathcal{U}_{h,\kappa} \to \mathcal{U}_h$ in every $L^p(d\mu_0)$.

Lemma C.2 (normalisation)**.** *For every $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$: $Z(h) \geq e^{-\lambda |a_0|\,|\Lambda|} > 0$, and $Z(h) < \infty$ by Theorem C.5. Hence dq_h is a well-defined probability measure on the whole class $\mathcal{C}_{\mathrm{E}}$.*

Proof. $\mathbb{E}[\mathcal{U}_h] = \lambda a_0 \int h \in [-\lambda|a_0||\Lambda|, \lambda|a_0||\Lambda|]$, since Wick monomials of degree ≥ 1 have mean zero and $0 \leq h \leq \mathbf{1}_\Lambda$. Jensen's inequality for the probability measure $d\mu_0$ gives $Z(h) \geq e^{-\mathbb{E}[\mathcal{U}_h]} \geq e^{-\lambda|a_0||\Lambda|}$. □

Lemma C.3 (Wick lower bound; the $h \geq 0$ entry)**.** *There is $b = b(\mathscr{P})$ such that for all $c \geq 0$ and $x \in \mathbb{R}$, $:\mathscr{P}:_c(x) := \sum_n a_n c^{n/2} \operatorname{He}_n(x/\sqrt{c}) \geq -b\,(1+c)^{p'}$. Consequently, for every $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$ and every $\kappa \geq 2$,*

$$\mathcal{U}_{h,\kappa} \;\geq\; -\lambda\, b\, |\Lambda| \left(1 + \tfrac{1}{2\pi} \log \kappa + C_*\right)^{p'} \;\geq\; -c_1(\lambda, \mathscr{P}, \Lambda)\, (\log \kappa)^{p'}.$$

Proof. Expanding the Hermite polynomials, $:\mathscr{P}:_c(x) = \sum_{m=0}^{2p'} \beta_m x^m$ with $\beta_{2p'} = a_{2p'}$ and $|\beta_m| \leq C(\mathscr{P})(1+c)^{(2p'-m)/2}$ (each power c^j comes with x^{n-2j}). For a polynomial $Ax^{2p'} + \sum_{m<2p'} \beta_m x^m$ with $A > 0$, weighted AM–GM gives, for each $m < 2p'$,

$$|\beta_m||x|^m \;\leq\; \frac{A}{4p'}\, x^{2p'} + C_{p'} \Big(\frac{|\beta_m|^{2p'}}{A^m} \Big)^{1/(2p'-m)},$$

and summing, $\min_x :\mathscr{P}:_c \geq -C'_{p'} \sum_m (|\beta_m|^{2p'}/A^m)^{1/(2p'-m)}$. With $|\beta_m| \leq C(1+c)^{(2p'-m)/2}$ the exponent of $(1+c)$ is $\frac{2p'}{2p'-m} \cdot \frac{2p'-m}{2} = p'$, uniformly in m. The second claim follows from $h \geq 0$, $\int h \leq |\Lambda|$, and the pointwise bound applied under the integral — this is the single use of the *sign* of h. □

Lemma C.4 (ultraviolet difference; the $h \leq 1$ entry)**.** *There is $c_2 = c_2(\lambda, \mathscr{P}, \Lambda)$ such that for all $h \in \mathcal{C}_{\mathrm{E}}$ with $\operatorname{supp} h \subseteq \Lambda$ and all $\kappa \geq 2$,*

$$\big\| \mathcal{U}_h - \mathcal{U}_{h,\kappa} \big\|^2_{L^2(d\mu_0)} \;\leq\; c_2\, \kappa^{-1} (\log \kappa)^{2p'}.$$

Proof. By the Wick isometry, with $G^{(1)}_\kappa := (\delta_\kappa \otimes 1) * G$ and $G_\kappa := (\delta_\kappa \otimes \delta_\kappa) * G$,

$$\|\mathcal{U}_h - \mathcal{U}_{h,\kappa}\|^2_{L^2} = \lambda^2 \sum_{n=1}^{2p'} a_n^2\, n! \iint h(z) h(z') \left[G^n - 2(G^{(1)}_\kappa)^n + G^n_\kappa \right](z - z')\, dz\, dz'$$

(the c_κ-Wick ordering makes the diagonal subtractions match; degree 0 cancels). Using $|h| \leq \mathbf{1}_\Lambda$ — the single use of the *upper* bound on h — it suffices to bound $\iint_{\Lambda\times\Lambda} |G^n - J^n|$ for $J \in \{G^{(1)}_\kappa, G_\kappa\}$. The classical Bessel facts $0 < G(\zeta) \leq C(1 + |\log|\zeta||) e^{-m_0|\zeta|}$ and $|\nabla G(\zeta)| \leq C(|\zeta|^{-1} + 1) e^{-m_0|\zeta|}$ give, for the average J of G over balls of radius $\leq 2/\kappa$: $J(\zeta) \leq C(1 + \log \kappa)$ for all ζ, and $|J(\zeta) - G(\zeta)| \leq C\kappa^{-1}(|\zeta|^{-1} + 1)$ for $|\zeta| \geq 4/\kappa$. Split the integral over $\zeta = z - z'$:

$$\int_{|\zeta|<4/\kappa} \big[G^n + J^n\big] \leq C_n \Big[\int_{|\zeta|<4/\kappa} (1 + |\log|\zeta||)^n d\zeta + \kappa^{-2} (1 + \log \kappa)^n \Big] \leq C'_n \kappa^{-2} (\log \kappa)^n,$$

$$\int_{4/\kappa \leq |\zeta| \leq R} n \max(G, J)^{n-1} |G - J| \leq C_n \kappa^{-1} \int_0^R (1 + |\log r|)^{n-1} (1 + r)\, dr \leq C_n(R)\, \kappa^{-1},$$

with $R = \operatorname{diam} \Lambda$; multiply by $|\Lambda|$ for the outer variable and sum over $n \leq 2p'$. □

Theorem C.5 (h-uniform stability)**.** *For every $p < \infty$ and compact Λ,*

$$\sup\{\|e^{-\mathcal{U}_h}\|_{L^p(d\mu_0)} \;:\; h \in \mathcal{C}_{\mathrm{E}},\ \operatorname{supp} h \subseteq \Lambda\} \;<\; \infty.$$

Proof. (For $\lambda = 0$, $\mathcal{U}_h = 0$ and the statement is trivial; assume $\lambda > 0$.) $Y := \mathcal{U}_h - \mathcal{U}_{h,\kappa}$ is a sum $Y = \sum_{r=0}^{2p'} Y_r$ of Wick chaoses, so Nelson's hypercontractive moment comparison applies in the form

$$\|Y\|_{L^q} \le (q-1)^{p'} \|Y\|_{L^2} \qquad (q \ge 2).$$

(Proof from the abstract form of Assumption A4 [12, Thm. I.17]: the second-quantized contraction $\Gamma(e^{-t})$ on $L^2(d\mu_0)$ satisfies $\|\Gamma(e^{-t})X\|_q \le \|X\|_2$ for $q = 1 + e^{2t}$, and $\Gamma(e^{-t})Y_r = e^{-rt}Y_r$. Write $Y = \Gamma(e^{-t})Z$ with $Z = \sum_r e^{rt}Y_r$; by orthogonality of the chaoses, $\|Z\|_2 \le e^{2p't}\|Y\|_2$, whence $\|Y\|_q \le \|Z\|_2 \le e^{2p't}\|Y\|_2 = (q-1)^{p'}\|Y\|_2$.)

For $b \ge b_0(\lambda, \mathscr{P}, \Lambda)$ choose $\kappa = \kappa(b)$ by $c_1(\log\kappa)^{p'} = b/2$, i.e. $L := \log\kappa = (b/2c_1)^{1/p'}$. By Lemma C.3, $\{\mathcal{U}_h \le -b\} \subseteq \{\mathcal{U}_h - \mathcal{U}_{h,\kappa} \le -b/2\}$, so by Chebyshev in L^{2q} and Lemma C.4,

$$\Pr[\mathcal{U}_h \le -b] \;\le\; \Big(\frac{2}{b}\Big)^{2q} (2q)^{2p'q} \left(c_2\, e^{-L} L^{2p'}\right)^q \;=\; \Big[\frac{2\sqrt{c_2}}{b}\,(2q)^{p'}\, L^{p'} e^{-L/2}\Big]^{2q}.$$

Choose $2q = \exp\big(\frac{L}{4p'}\big)$, so that $(2q)^{p'} = e^{L/4}$ and the bracket is $\le \frac{2\sqrt{c_2}}{b} L^{p'} e^{-L/4} \le e^{-1}$ for $b \ge b_1(\lambda, \mathscr{P}, \Lambda)$, whence

$$\Pr[\mathcal{U}_h \le -b] \;\le\; \exp\Big(-\exp\Big(\tfrac{1}{4p'}(b/2c_1)^{1/p'}\Big)\Big),$$

doubly exponentially small, uniformly over the stated class; and $\exp\big(\frac{1}{4p'}(b/2c_1)^{1/p'}\big) \ge 2pb$ for b large, since a power of b inside an exponential beats $\log b$. Then $\|e^{-\mathcal{U}_h}\|_{L^p}^p = \int_0^\infty p e^{pb} \Pr[\mathcal{U}_h \le -b]\, db + O(e^{pb_1}) < \infty$, with a bound depending only on $(p, \lambda, \mathscr{P}, \Lambda)$. Every constant above came from Lemmas C.3–C.4, which used exactly $h \ge 0$ and $h \le 1$. □

Remark C.6 (why the class is what it is)**.** The two halves of the constraint $0 \le h \le 1$ are used in complementary places: positivity in the pointwise lower bound (Lemma C.3 would fail for h of variable sign — the interaction density is unbounded above *and* below), and boundedness in the L^2 kernel estimates (Lemma C.4 and M3). Nothing else about h — smoothness, shape, time extent, monotonicity along a sequence — is touched in Lemmas C.3–C.4 or in M3; the stability input of the source's own proof (M6) is a separate matter. This is the structural content of Source Fact 2.4.

Proposition C.7 (coverage of every CE application)**.** *Every use of CE1 in §§4–7 has a cutoff in $\mathcal{C}_{\mathrm{E}}^{\mathrm{slab}}$ and is covered by (U_{GJS}) (Assumption A8(vi), the printed Theorem 1.1.11 at $B = I$, $n = 0$); every use of CE2 is covered directly by Source Fact 2.2, whose constants are independent of h, across strips parallel to the time-zero line, the case its printed proof reaches. Lemma C.2 and Theorem C.5 supply the normalization and integrability needed to define the same measures. Hence no stronger cutoff-uniformity assertion is consumed by the proof.*

Proof. The CE1 occurrences are the displays in Theorem 5.1, Lemmas 7.1–7.3, and their immediate consequences; in each case the measure is explicitly $dq_{h_{T,g}}$. If g is replaced by an affine path cutoff g_s, Lemma 3.1 gives $g_s \in \mathcal{C}_{\mathrm{H}}$, so $h_{T,g_s} \in \mathcal{C}_{\mathrm{E}}^{\mathrm{slab}}$. Lemma 7.4 is the only connected CE2 application and Source Fact 2.2 quantifies over its cutoff directly. The remaining assertions are exactly Lemma C.2 and Theorem C.5. □

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